AD Server, DNS Server, Domain Controller, Server Data Sync & DNS Architecture

Introduction to Directory Services

A Directory Service is best understood as a digital phonebook, or a central database, for an entire computer network.

To understand why this is needed, imagine a large company with hundreds of users, computers, printers, and shared folders. Keeping track of all of these things separately, on every single computer, would be extremely difficult to manage. A directory service solves this problem by storing information about all of these resources in one single, central place, and then helping to manage and control them from there.

In simple terms, here is what a directory service actually does:

- **Central Storage:** It keeps an organized, structured list of every user, password, and device that exists within the network.
- **Log-in Control (Authentication):** Whenever a user sits down at any computer in the network and types in their username and password, that computer asks the Directory Service a simple question: "Is this password correct, and should this person be allowed in?"
- **Access Control (Authorization):** It defines exactly who is allowed to do what, within the network. For example, it can enforce a rule such as: "Only allow the Finance department to open the Finance folder."

Most companies around the world use Microsoft Active Directory as their main directory service. However, it is worth knowing that other directory services also exist, such as LDAP, OpenLDAP, and Novell eDirectory.

DNS Server

A DNS (Domain Name System) server can be thought of as the phonebook of the Internet.

Here is why this is necessary: computers and routers do not actually understand website names like google.com or facebook.com. Instead, they only understand numbers, known as IP addresses (for example, 142.250.190.46).

The job of a DNS server is to translate these human-friendly website names into computer-friendly IP addresses, so that our web browser is able to correctly load the page we are trying to visit.

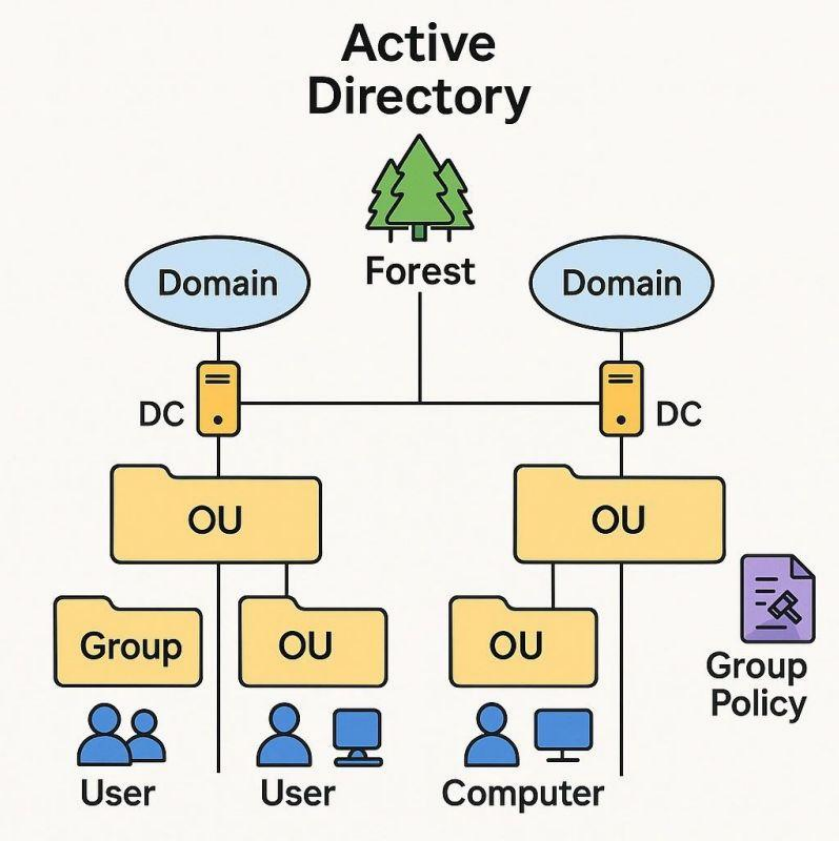
Active Directory (AD) Architecture

Active Directory is organized using a layered, logical structure. This structure helps organize all the different resources of a network, based on things like company departments or geographical office locations, completely separately from the actual physical hardware being used.

1. The Logical Architecture Layer

This layer is purely conceptual, meaning it exists as an organizational idea, rather than as physical hardware. It is made up of several building blocks, which fit together as follows:

- **Objects:** These are the smallest and most basic building blocks inside Active Directory. An object represents one specific resource, such as a User, a Computer, a Printer, or a Shared Folder.
- **Organizational Units (OUs):** These are containers, used to group similar objects together within a domain. For example, we could create an OU named "Finance", and place all of the finance department's user accounts and computers inside it. This makes it much easier to apply specific security rules, known as Group Policies, to just that one department, without affecting anyone else.
- **Domains:** This is the core logical boundary within an Active Directory network. A domain shares one common database, one common security policy, and one common name (for example, company.local). All objects that exist inside the same domain are managed together, as one unit.
- **Trees:** A tree is a collection of one or more domains, which all share a continuous naming structure. For example, uk.company.local and us.company.local could both be child domains, under one shared parent domain called company.local. Together, these domains form what is called a Tree.
- **Forests:** This is the topmost layer of the entire logical structure. A forest is a collection of one or more independent Trees, which all share a common global catalog, a common directory schema, and a common overall configuration. The forest represents the ultimate, outermost security boundary of an entire organization.



The diagram above helps visualize this whole structure: the Forest sits at the very top, containing one or more Domains. Each Domain is managed by its own Domain Controller (DC). Inside each domain, Organizational Units (OUs) are used to organize Users, Computers, and Groups, and Group Policies are applied to control and secure these objects.

Domain Controller (DC)

A Domain Controller, usually just called a DC, is the single most important server within a Microsoft Active Directory network. It effectively acts as the "brain" and the central security hub of the entire company's network.

Core Responsibilities of a Domain Controller

- **User Authentication:** Whenever we type our username and password on any computer within the office, the Domain Controller checks this information against its own database, to verify who we are, before logging us in.
- **Access Authorization:** The Domain Controller also decides exactly what we are allowed to see and do, once we are logged in. For example, it might allow the Finance team full access to sensitive financial files, while blocking regular staff members from ever accessing the IT department's servers.
- **Central Database:** It stores one master database file, called ntds.dit, which contains all of the network's important data, including user accounts, passwords, computer records, and group permissions.
- **Rule Enforcement (Group Policy):** It allows IT administrators to push out security rules to every single computer in the network at once. For example, an administrator could use this to block all USB drives across the company, or to force every user to change their password every 30 days.

Global Catalog (GC)

The Global Catalog can be described as a searchable, partial replica (a partial copy) of all the objects that exist across every domain in the forest, kept in one place for fast, efficient searching.

In simple terms, the Global Catalog acts as the master search engine of an entire Active Directory Forest. It holds a small, condensed copy of everything important in the network, so that users can instantly find any resource they need, without slowing the network down.

1. Searchable Partial Replica

The word "partial" is important here. The Global Catalog does not copy absolutely everything from every other domain, because doing so would take up far too much storage space, and would slow the server down significantly. Instead, it only copies a smaller, carefully chosen set of the most important information needed for searching, such as a user's first name, last name, email address, and logon name.

2. Working Across Domains

In a large company that has multiple departments or branches, for example sales.company.com, it.company.com, and hr.company.com, a normal, standard server usually only knows about the information within its own branch. The Global Catalog breaks down these barriers, and instead holds combined information from all of these different branches together, in one place.

3. Efficient Queries

A "query" is simply another word for a search request. Because the Global Catalog keeps one master index of information in a single, central place, our computer does not need to waste time asking every

single individual server in the company, one by one. Instead, it gets its answer almost instantly, which is what makes searching across the network fast and efficient.

AD Authentication: How Users Log In Safely

Whenever we type in our password to log in, Active Directory relies on two main security systems, to properly verify exactly who we are:

- **Kerberos v5:** This is the modern, and highly secure, standard authentication system used today. It works using digital "tickets." Once we log in successfully, the server issues us a ticket, which then allows us to open shared folders and access other resources, without needing to type in our password over and over again each time.
- **NTLM:** This is an older authentication system. It is considerably less secure than Kerberos, but it is still kept active within many networks, so that very old computers, or legacy corporate software still in use in the office, are still able to connect to the network, without breaking their existing functionality.

AD Replication: How Servers Sync Their Data

If a company has more than one Domain Controller (meaning more than one server acting in this role), all of these Domain Controllers need to have exactly the same, matching data at all times.

- **Multi-master replication:** This means that any single Domain Controller server is able to accept new changes, such as a password reset, and it will then automatically sync and copy that new data across to all of the other Domain Controllers in the network.
- **Intra-site (within the same LAN):** This refers to copying data between servers that are located inside the same office. Since these servers are connected through a fast local area network (LAN), the data syncs almost instantly between them, without even needing to be compressed first.
- **Inter-site (across a WAN):** This refers to copying data between servers that are located in different branches or different cities. Since WAN or Internet connections between these distant locations can often be much slower, Active Directory compresses this data first (making it smaller), and then syncs it according to a set schedule, in order to save valuable network bandwidth.

FSMO Roles: Acting as an Additional Domain Controller

Normally, if an Active Directory network has multiple Domain Controllers, they all share equal power and responsibility between them; this is known as multi-master management.

However, there are 5 highly sensitive tasks within the network, which absolutely cannot afford to have any conflicts, or any overlapping mistakes between different servers. Because of this, instead of allowing every single Domain Controller to handle these particular tasks at the same time, these 5 critical responsibilities are instead assigned exclusively to specific, individual servers. These 5 special roles are collectively known as the FSMO Roles, which stands for Flexible Single Master Operations.

- **Schema Master:** Manages the overall blueprint of Active Directory. This role applies across the whole forest.

- **Domain Naming Master:** Controls the process of adding or removing entire domains. This role also applies across the whole forest.
- **RID Master:** Assigns unique ID number pools to Domain Controllers, which they then use to create new objects. This role applies at the domain level.
- **PDC Emulator:** Handles time synchronization across the network, password changes, and Group Policy processing. This role applies at the domain level.
- **Infrastructure Master:** Synchronizes object information correctly across different domains. This role also applies at the domain level.

DNS Concept & Architecture

For a helpful video explanation of how a DNS server actually works, see this reference: How DNS Server Works (YouTube)

1. The Main DNS Server Inside a Company (Local DNS)

Most companies do not rely on external or public DNS servers as their main, primary choice. Instead, they choose to run their own DNS service directly inside their Active Directory Domain Controller (DC).

Whenever any computer on the company network automatically receives its IP configuration through DHCP, the IP address of this same Active Directory server is also assigned to it, to be used as that computer's primary DNS server.

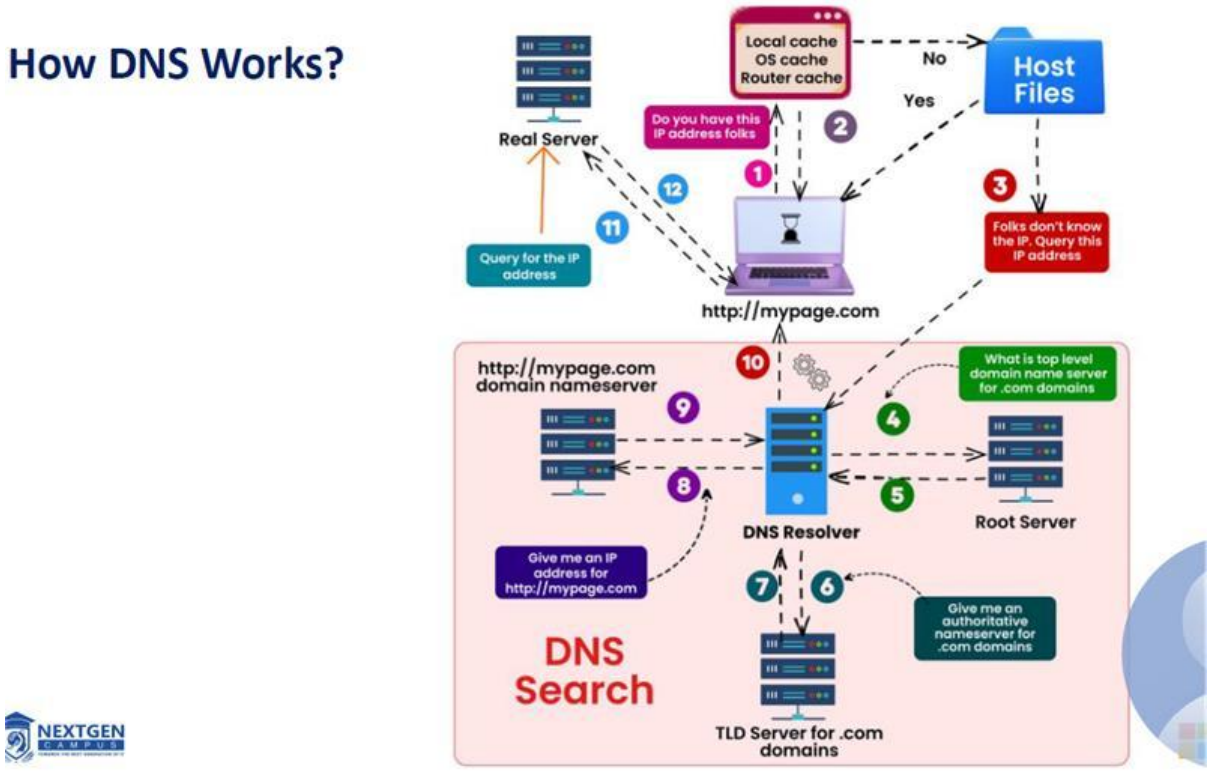
2. Searching for an Internal Website or Server (Internal Query)

Let's imagine we are working in the IT department, and we want to access the company's internal HR portal, at an address like hr.company.local.

- Our computer asks the internal DNS server directly: "Where is hr.company.local located?"
- This internal DNS server keeps a complete list of IP addresses, for every single PC and server in the company, stored inside its own database. This database is known as the Forward Lookup Zone.
- The server then instantly returns the exact matching IP address (for example: 192.168.20.5) back to our computer, and the internal website opens up successfully for us.

How DNS Works (Full Process)

When we type a website address into our browser, a longer background process actually takes place, involving several different servers working together, before the page finally loads.



As shown in the diagram above, here is a simplified version of what happens, step by step:

- First, our computer checks its own local cache, OS cache, and router cache, to see if it already knows the IP address for the website we typed in. If it already knows the answer, it uses that immediately, without going any further.
- If our computer does not already know the answer, it checks a local list called the Host Files. If that also does not contain the answer, our computer sends the request onward to a DNS Resolver.
- The DNS Resolver then works through a series of servers to find the answer: it may ask a Root Server, which points it toward the correct Top Level Domain (TLD) Server (for example, the server responsible for all ".com" domains).
- The TLD Server then points the resolver toward the correct authoritative domain nameserver, which is the specific server that actually knows the exact IP address for the website being requested.
- Once the DNS Resolver receives this final IP address, it passes that information back to our computer.
- Finally, our computer uses this IP address to contact the real web server directly, and the website is loaded in our browser.

Even though this entire process sounds long, it actually all happens within a very small fraction of a second, in the background, every single time we visit a new website.

DNS Zones

- **Primary Zone:** This zone holds the master, original copy of all DNS records, and it supports both reading and writing (making changes) to those records.
- **Secondary Zone:** This is a read-only copy of the DNS records, which is regularly updated from the Primary Zone, through a process called zone transfer.

- **Stub Zone:** This zone contains only NS (Name Server) records, and its main purpose is to help resolve names correctly across different domains.
- **Forward Lookup Zone:** This zone is used to map a website or hostname to its correct IP address (Name to IP).
- **Reverse Lookup Zone:** This zone works in the opposite direction, mapping an IP address back to its correct hostname (IP to Name).

DNS Record Types

- **A Record:** Maps a hostname directly to an IPv4 address. For example: app.bank.lk → 192.168.1.10
- **AAAA Record:** Maps a hostname directly to an IPv6 address. For example: server.bank.lk → 2001:db8::5
- **CNAME Record:** Acts as an alias, pointing to another existing record. For example, mail.bank.lk might actually just be a link that points to the main name, exchange.bank.lk.
- **PTR Record:** Used for reverse lookups. For example: 192.168.1.10 → app.bank.lk
- **MX Record:** Identifies the Mail Exchange server responsible for handling email. For example, mail.bank.lk is the server that is told to handle incoming mail whenever someone sends an email to info@bank.lk.
- **SRV Record:** Helps Active Directory locate specific Domain Controllers and services, such as LDAP. This record type is extremely important for Active Directory, because whenever a computer logs into the network, it reads this SRV record specifically, in order to find out exactly where the Domain Controller (DC) is located.

Implementing AD Server and Client in Hyper-V: Creating OUs, Users, and Connecting a Client to AD, Then Assigning a Wallpaper Policy

Active Directory: First Important Step

When we are setting up Active Directory for the very first time, there is an important security step we must do right away: we should either rename the built-in "Administrator" account, or completely disable it, and always use a strong, complex password. This is important because the Active Directory administrator account has the highest level of privilege in the entire network, meaning it has full control over everything.

Why rename or disable the Administrator account?

Every single Windows Server comes with a built-in account, which is automatically named "Administrator." Since this name is always the same by default, and well known to attackers, hackers can easily guess that this account exists, and then try to attack it directly. By either renaming this account to something less obvious, or disabling it completely and using a different account instead, we make it much harder for hackers to guess our login details.

Why an Internal Switch Was Chosen for the Active Directory Lab

When we set up Active Directory inside a virtual environment using Hyper-V, choosing the correct type of virtual switch is extremely important, for both network isolation and overall stability. For this particular project, an Internal Switch was chosen, instead of an External Switch, a Private Switch, or the Default Switch. Here is the reasoning behind that choice, explained in detail:

1. Isolation from the Production Network (compared to an External Switch)

An Active Directory environment needs its own dedicated infrastructure services, such as DHCP and DNS, running specifically for the lab. If we were to use an External Switch instead, our virtual Domain Controller would actually be connected to the same physical network as our real router and other real devices. This could cause serious conflicts, such as IP address duplication, and it could even disrupt the internet connection for other real devices on the same physical network.

By using an Internal Switch instead, our entire lab environment is completely isolated from the physical, real-world network, which means we can safely experiment and test things, without any risk of interfering with other devices.

2. Host-to-VM Communication (compared to a Private Switch)

A Private Switch does provide 100% isolation, meaning it is extremely secure. However, this also means that it completely blocks the physical host computer (our actual laptop) from being able to communicate with the virtual machines at all.

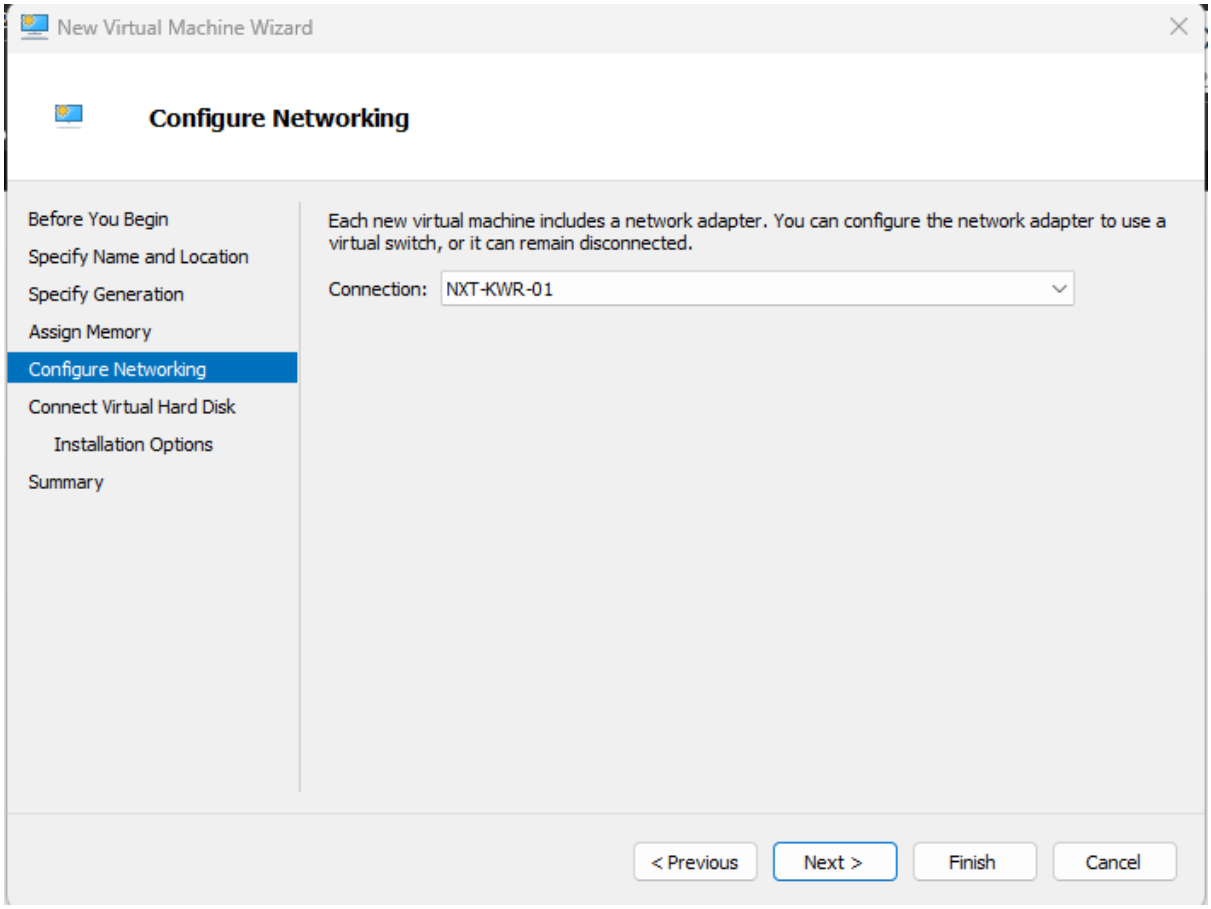
An Internal Switch, on the other hand, still allows the host computer to communicate with the virtual machines, even though the lab remains isolated from the outside network. This makes it much easier for us to transfer installation files, ISO images, and other useful tools directly from our laptop into the virtual servers, which would not be possible with a fully isolated Private Switch.

3. IP Address Stability (compared to the Default Switch)

The Hyper-V Default Switch automatically assigns dynamic IP addresses to virtual machines, and these IP addresses can actually change every single time the host computer restarts.

However, Active Directory and Domain Controllers strictly require a fixed, permanent Static IP address, in order to function correctly and reliably. If the IP address were to change unexpectedly after a restart, it could break the entire Active Directory setup.

By using an Internal Switch instead, we are able to manually configure a permanent, stable Static IP address for our lab environment, one that will not change or break, even after the host computer is restarted.

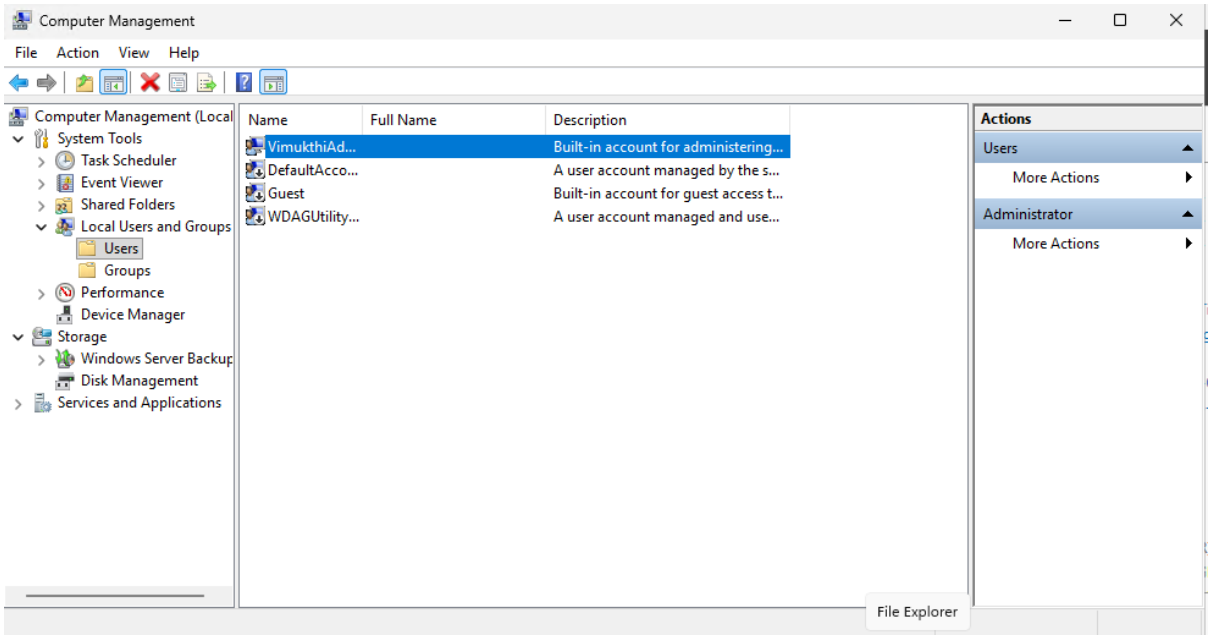


How to Rename the Administrator Username

To rename the Administrator account on our server, we can follow this simple path:

1. Open **Tools**.
2. Go to **Computer Management**.
3. Select **Local Users and Groups**.
4. Click on **Users**.

Once we are inside the Users folder, we can find the built-in "Administrator" account in the list, right-click on it, and rename it to something less obvious, so it becomes harder for anyone to guess.

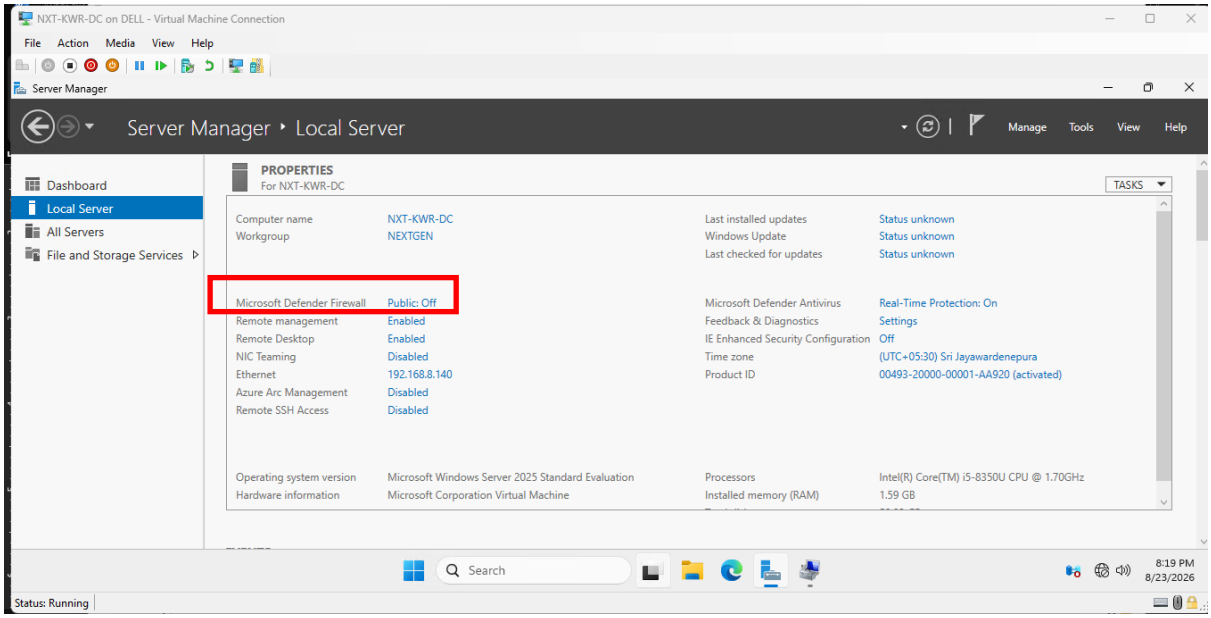


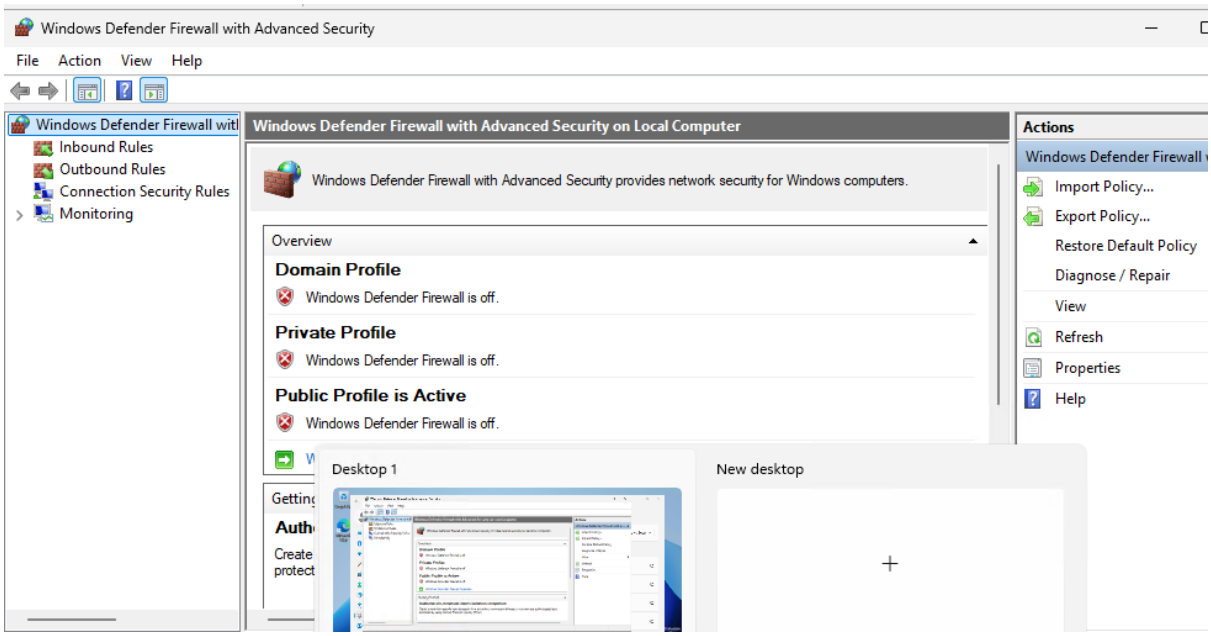
Firewall Considerations: Real World vs Lab Environment

In a real-world production environment, the firewall should always be enabled. In that situation, we need to carefully follow best practices regarding ports, meaning we must decide exactly which ports need to be opened (enabled), and which ports should remain closed (disabled), based on what services the server actually needs to run.

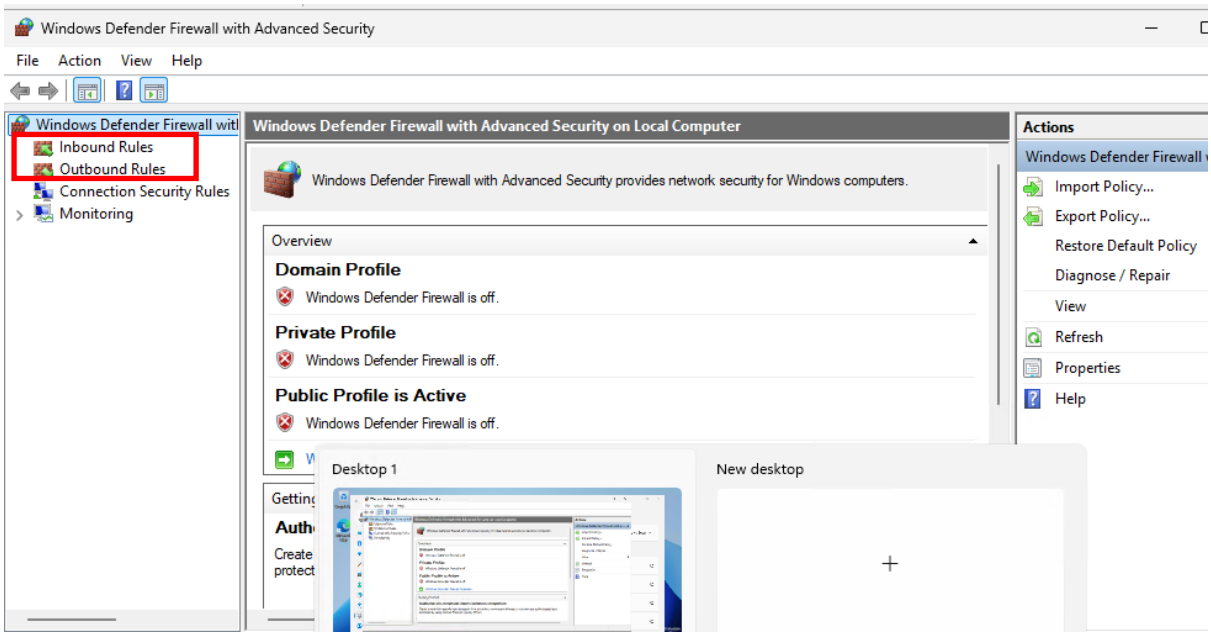
However, in this particular lab session, since we are only practicing and experimenting in a safe, isolated environment, we simply disable the firewall completely. This makes it easier for us to focus on learning Active Directory itself, without needing to also configure detailed firewall rules at the same time.

Note: Disabling the firewall is acceptable for lab and practice purposes, but this should never be done in a real, production server environment, as it would create serious security risks.





Those are the inbound rules and outbound rules



Adding a New Firewall Rule

If we want to add a new inbound rule (controlling traffic coming into the computer) or a new outbound rule (controlling traffic leaving the computer), we can simply click on the "New Rule" option, found on the right-hand side of the Windows Firewall management screen. This allows us to manually create a custom rule.

Understanding Windows Firewall Profiles

Windows Firewall uses three separate network profiles, called Domain, Private, and Public. These profiles allow the firewall to automatically adjust its security level, depending on how trustworthy the currently connected network is considered to be.

1. Domain Profile

This profile becomes active whenever a computer is connected to a network that is managed by a corporate network's domain controller, meaning it is part of an Active Directory environment.

- **Trust Level:** Highest trust.
- **Behavior:** Since the network is assumed to be secure and properly managed by IT administrators, the firewall allows corporate management traffic, background services, and file sharing to run smoothly, without unnecessary restrictions.

2. Private Profile

This profile is intended for home networks, or trusted small business networks, where the user themselves has full control over which devices are connected.

- **Trust Level:** Medium trust.
- **Behavior:** This profile enables Network Discovery, which allows our computer to see, and be seen by, other trusted devices on the same network, such as home printers or shared storage drives.

3. Public Profile

This profile is used by default whenever we connect to an untrusted public network, such as Wi-Fi hotspots found at airports, hotels, or coffee shops.

- **Trust Level:** Lowest trust (considered untrusted).
- **Behavior:** The firewall switches into maximum security mode. It completely disables Network Discovery, and it blocks other computers connected to the same public Wi-Fi network from being able to discover or access our device in any way.

Default Behavior vs Custom Rules

Whenever we install a new server role, such as Active Directory or DNS, onto a Windows Server, the operating system automatically creates and enables a set of basic, pre-defined firewall rules, which are specifically required for that particular service to function correctly.

However, even with these automatic rules already in place, system administrators still often need to manually create additional custom rules, for the following reasons:

1. Hardening Security

Custom rules can be used to restrict access even further, based on specific IP addresses or specific networks. For example, an administrator might configure a rule that only allows the IT Department's own subnet to access the server through Remote Desktop, blocking everyone else from doing so.

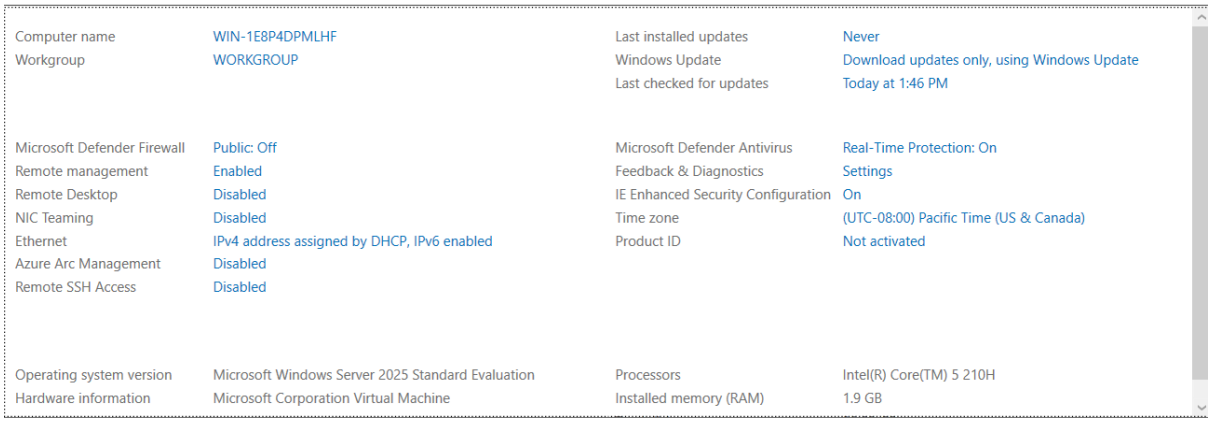
2. Supporting Third-Party Applications

Custom rules are also needed to allow network traffic for non-Microsoft software that the company might be using, such as accounting databases, HR systems, or backup tools. Since these are not built-in Windows services, they do not automatically receive their own pre-defined firewall rules, so an administrator must create one manually.

3. Network Troubleshooting

Sometimes, custom rules are needed simply for troubleshooting purposes. For example, a protocol like ICMP (commonly known as "Ping") is actually blocked by the firewall by default. An administrator may need to manually create a rule to allow ICMP traffic, so that they are able to properly test and confirm network connectivity between different clients and servers.

Server Basic Settings Overview



This screenshot shows the basic settings screen of a Windows Server, right after installation. It gives us a quick summary of important settings, such as the computer name, workgroup, firewall status, remote access settings, time zone, and hardware information, all in one place.

Remote Desktop (RDP) vs Remote Management

There are two common ways for an administrator to remotely access and control a server: Remote Desktop, and Remote Management. Even though both allow remote access, they work very differently, and are used for different purposes.

1. Remote Desktop (RDP)

Remote Desktop provides a full Graphical User Interface (GUI) of the remote server, meaning we see and interact with the server's actual desktop screen, exactly as if we were physically sitting in front of that machine ourselves.

- **Primary Use:** This is mainly used for manual configuration tasks, running graphical (GUI-based) applications, and visually troubleshooting problems on the server.
- **Default Port:** TCP port 3389.

2. Remote Management

Remote Management, on the other hand, allows administrators to manage and configure a server entirely in the background, without ever opening a full visual desktop session. Instead, it connects using command-line tools, such as PowerShell, or through remote management consoles, such as Server Manager or Windows Admin Center.

- **Primary Use:** This is mainly used for automation tasks, running scripts across many servers at the same time, and monitoring the overall health of a server, without using up heavy network bandwidth.
- **Default Ports:** TCP port 5985 (for HTTP) and TCP port 5986 (for HTTPS), both used through a system called WinRM (Windows Remote Management).

What is an NTP Server?

NTP stands for Network Time Protocol. An NTP Server is a specialized type of server, whose job is to synchronize the clocks of all computers, servers, and network devices within a network, making sure they all display the exact same, correct time.

Why is NTP so important for Active Directory?

Within a Windows Domain environment, having accurate, synchronized time across all devices is absolutely mandatory, for both security and authentication purposes.

- Windows uses a protocol called Kerberos for its authentication process.
- If there is a noticeable time difference, known as "Time Skew," between the Domain Controller (DC) and a client PC, of more than 5 minutes, then authentication will actually fail completely, and users will not be able to log into their computers at all.
- Because of this strict requirement, the Active Directory Domain Controller automatically takes on the role of the NTP Server for the entire local network, making sure that all client devices remain perfectly synchronized with each other at all times.\

How to connect AD via remote desktop (RDP)

To connect to our Active Directory server remotely, using Remote Desktop, we follow these steps:

1. First, we need to know the IP address of the Active Directory server.
2. Then, on the host computer (our laptop), we open the Remote Desktop Connection application, and type in that IP address.
3. We then enter the correct credentials, meaning the username and password of an account with access to that server.
4. We click "Yes" if a security certificate warning appears, and we are now successfully connected to the Active Directory server through RDP.

Important: Understanding APIPA Addresses

When we do not manually assign a static IP address to our server, Windows will automatically assign a temporary IP address instead, from a special range starting with 169.254.x.x. This type of automatically assigned address is called an APIPA address, which stands for Automatic Private IP Addressing.

Why does this happen with an Internal Switch?

When we use an Internal Switch in Hyper-V, there is no DHCP server available in that environment, to properly hand out real IP addresses. Because of this, Windows automatically creates its own temporary APIPA address instead, just so the network adapter has some kind of address to use.

Why is an APIPA address a problem for Active Directory?

An Active Directory server acts as the central "hub" of the entire network. However, an APIPA address is not stable, it can actually change unexpectedly, for example during a server restart, or due to other events. If this were to happen, client computers would suddenly be unable to find the server

anymore, and the entire network would effectively go down, since none of the clients would know the new address to connect to.

What happens if we use an External Switch instead?

If we use an External Switch instead of an Internal Switch, our virtual machine would be able to reach our actual physical router, and the router itself could hand out an IP address through DHCP. In that case, the APIPA issue would not occur, since a real IP address would already be available.

So why do we still need a Static IP address?

Even though using an External Switch with DHCP avoids the APIPA problem, the IP address received through DHCP from the router can still technically change at any time, since DHCP addresses are dynamic by nature.

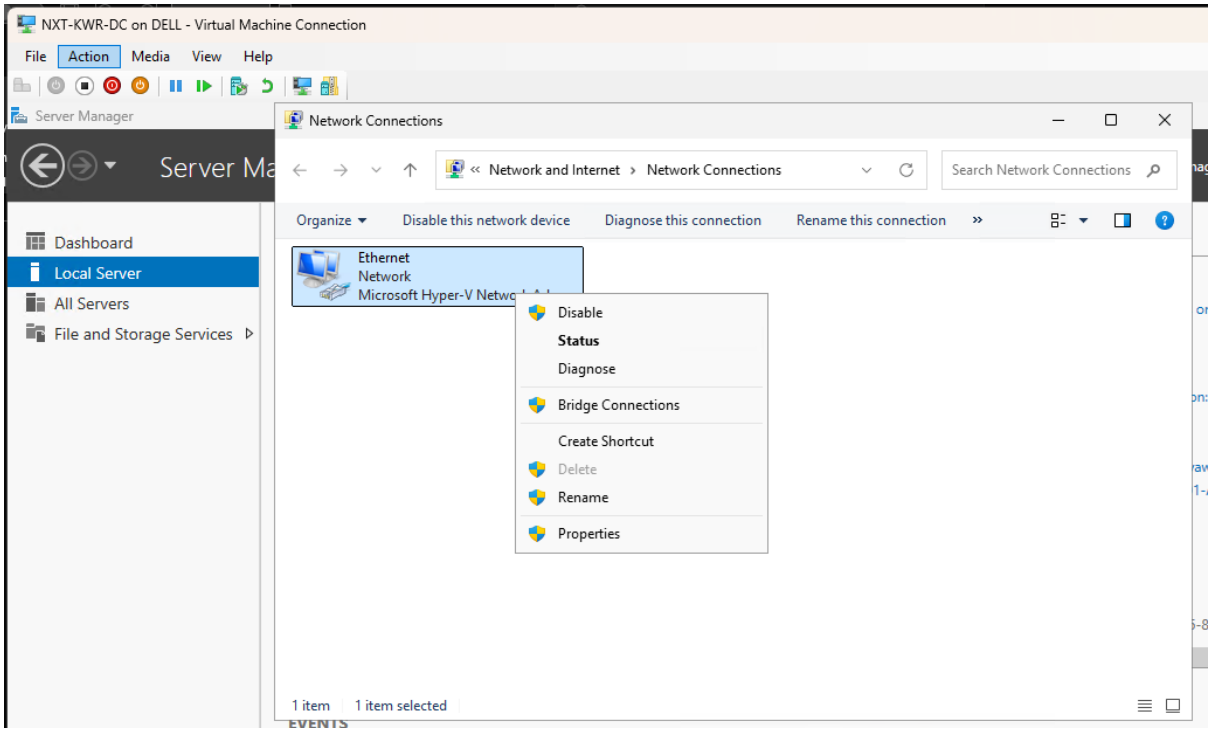
However, since an Active Directory server absolutely needs to keep the exact same address at all times, in order to remain reliable, we must manually assign it a Static IP address instead, one that will never automatically change, no matter what happens.

Assigning a Static IP Address to Our Server

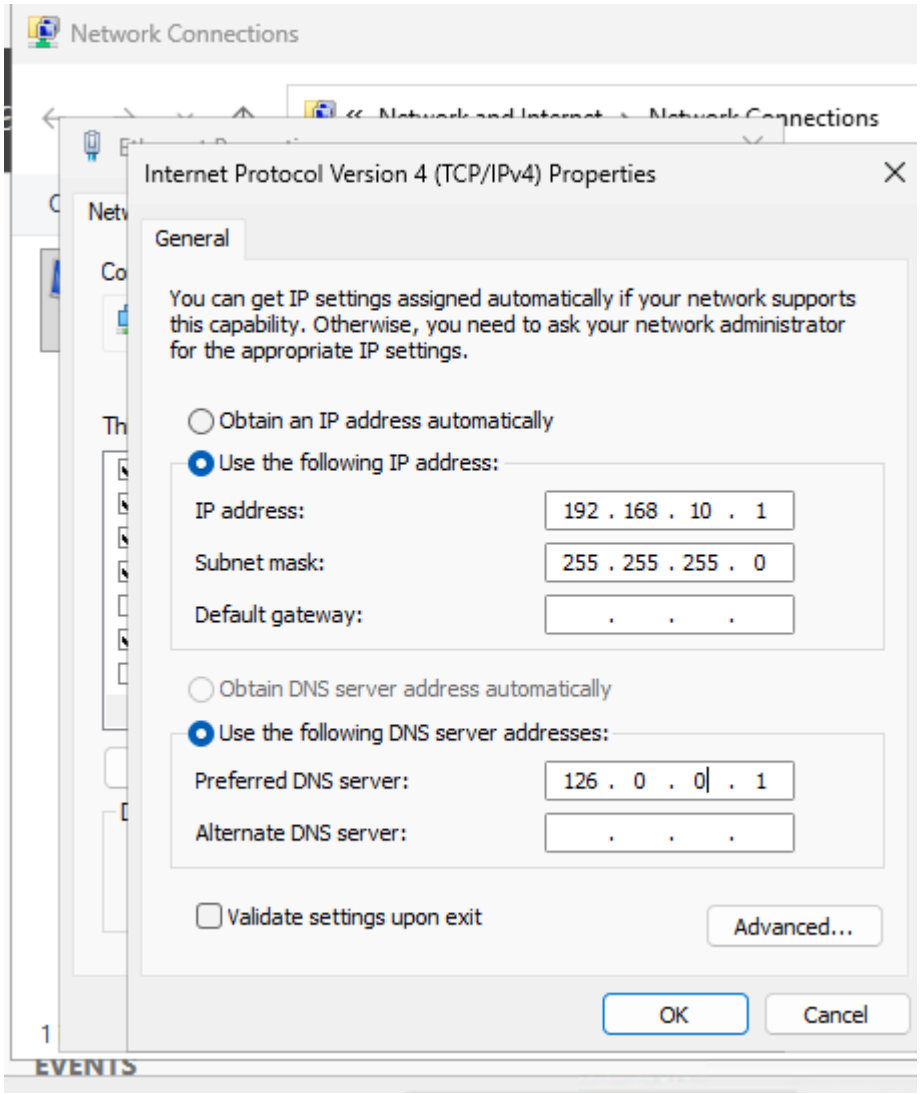
To assign a static IP address, we can follow this path:

Control Panel → Network and Internet → Network Connections

Alternatively, there is a faster shortcut available: press **Windows key + R**, then type **ncpa.cpl**, and press Enter. This will take us directly to the Network Connections screen, without needing to navigate through the Control Panel manually.



Select internet **Internet Protocol Version 4 (TCP/IPv4) and go properties**



Assigning a Static IP Address (Example Configuration)

In the screenshot above, we can see the Internet Protocol Version 4 (TCP/IPv4) Properties window, which is where we manually configure a static IP address on our server.

Here, the option "Use the following IP address" is selected, and the following details are entered:

- **IP address:** 192.168.10.1
- **Subnet mask:** 255.255.255.0
- **Preferred DNS server:** 127.0.0.1

Why We Use 127.0.0.1 for DNS Settings

The IP address 127.0.0.1 is known as the Loopback Address, or more commonly, "Localhost." This special address always refers back to the same machine itself, no matter which computer is using it.

The Role of DNS in Active Directory

Active Directory cannot function properly without DNS. This is because the DNS server is responsible for storing the exact locations of all domain services within the network. Because of this

close relationship, whenever we promote a server to become a Domain Controller, the DNS Server role is also automatically installed on that very same machine, alongside Active Directory.

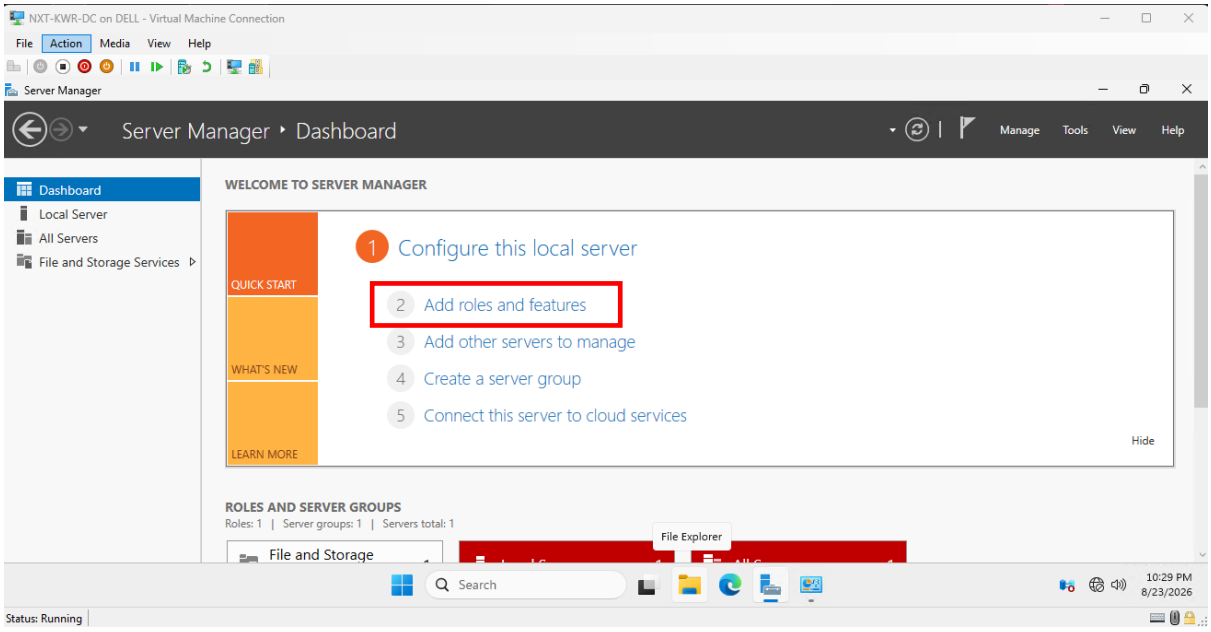
Why Do We Use the Loopback Address (127.0.0.1) for DNS?

1. Self-Resolution

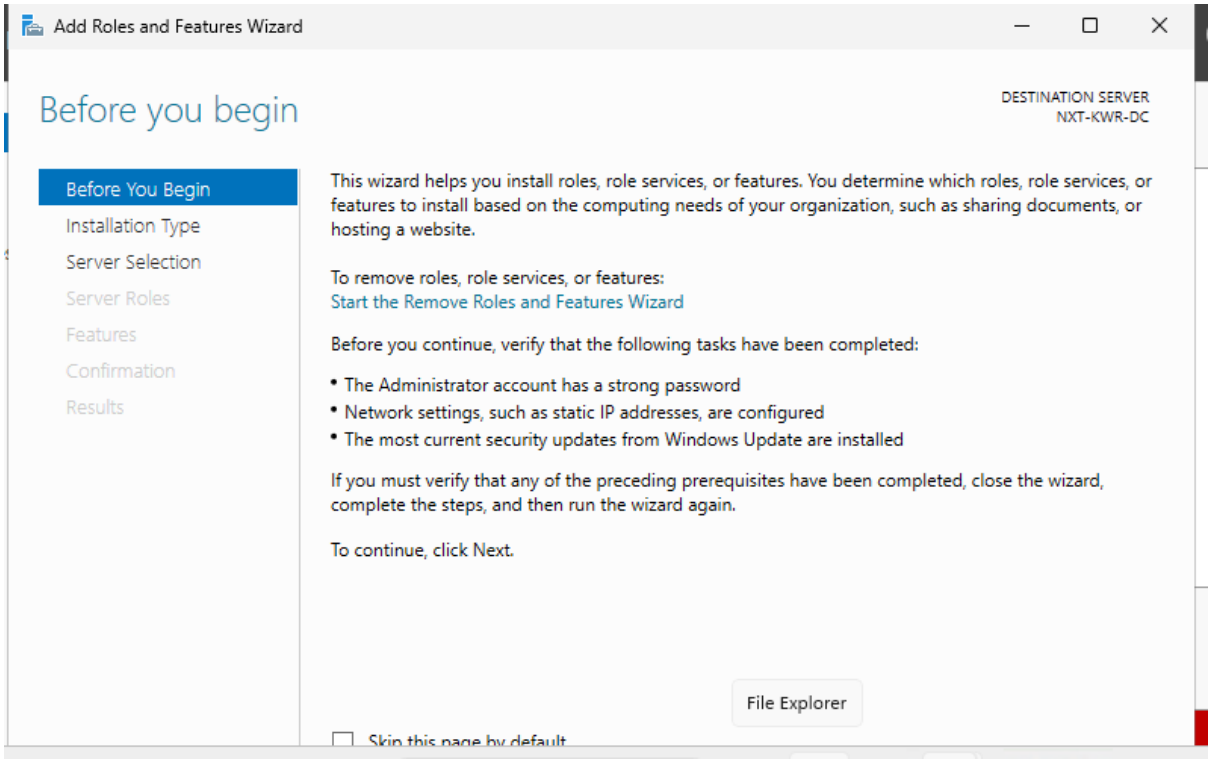
By setting our server's own DNS server address to 127.0.0.1, we are essentially telling Windows Server: "Use your own local DNS database, to resolve any DNS queries you receive." Since the DNS Server role is already installed directly on this same machine, the server does not need to look anywhere else for DNS information, it can simply refer to itself, using the loopback address, since it already contains its own complete DNS records.

This setup makes the server more reliable and self-sufficient, since it does not depend on any other external DNS server, in order to resolve its own domain-related queries.

How to add Active directory in server manager



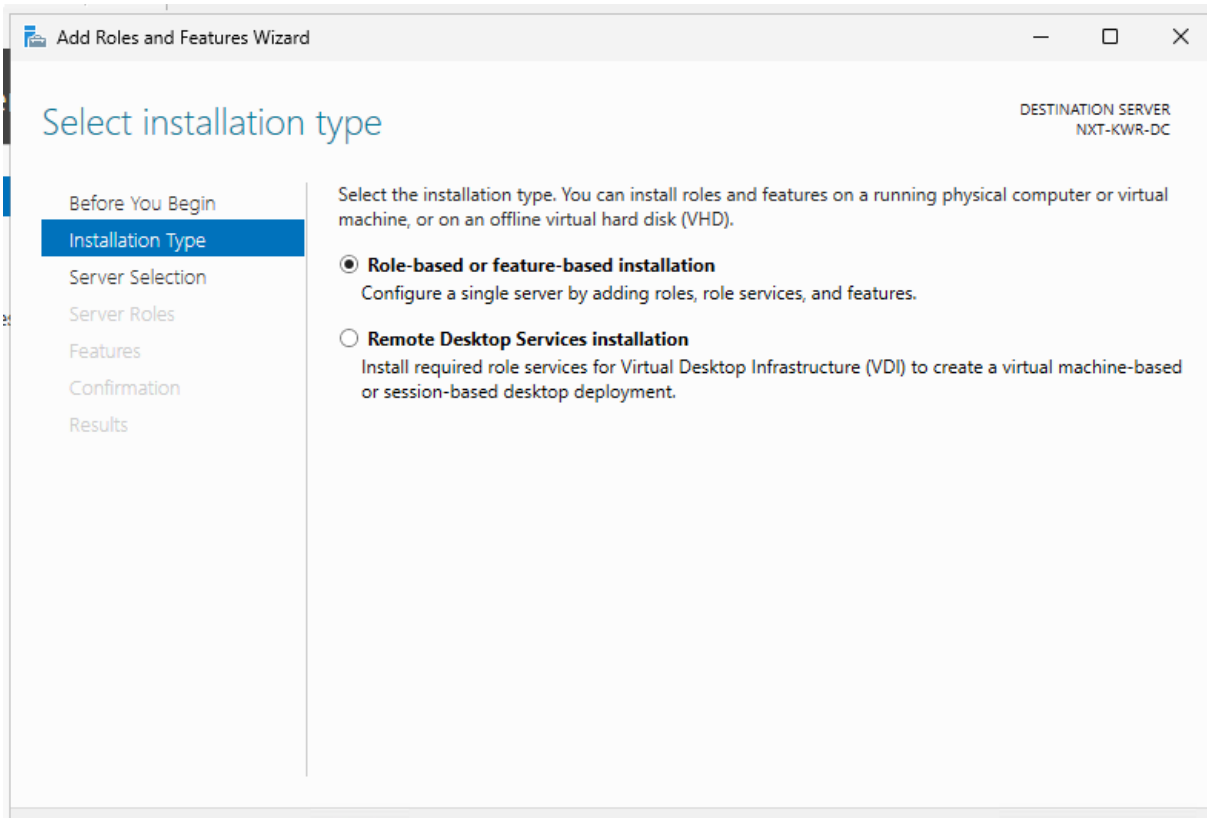
When you click the Add roles and Features section now you can see below options

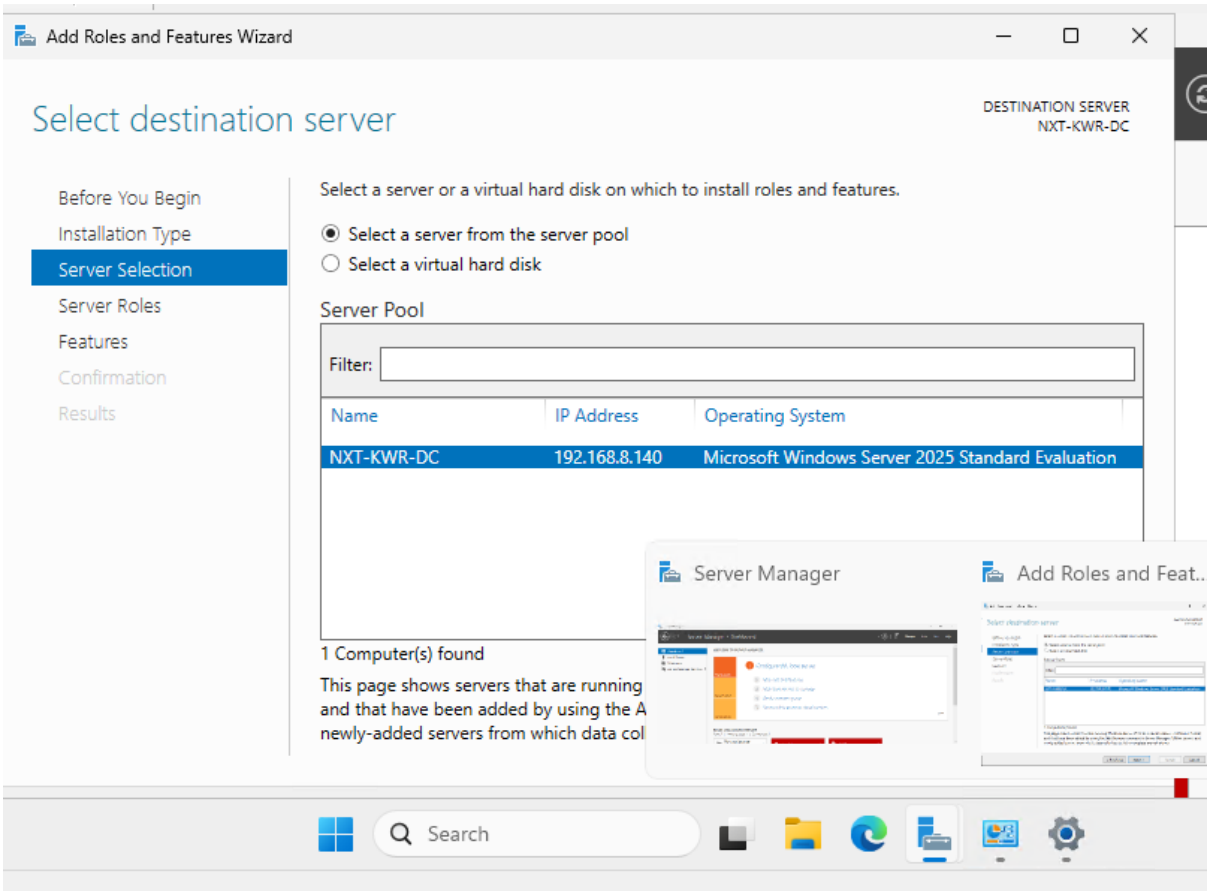


Selecting the Installation Type

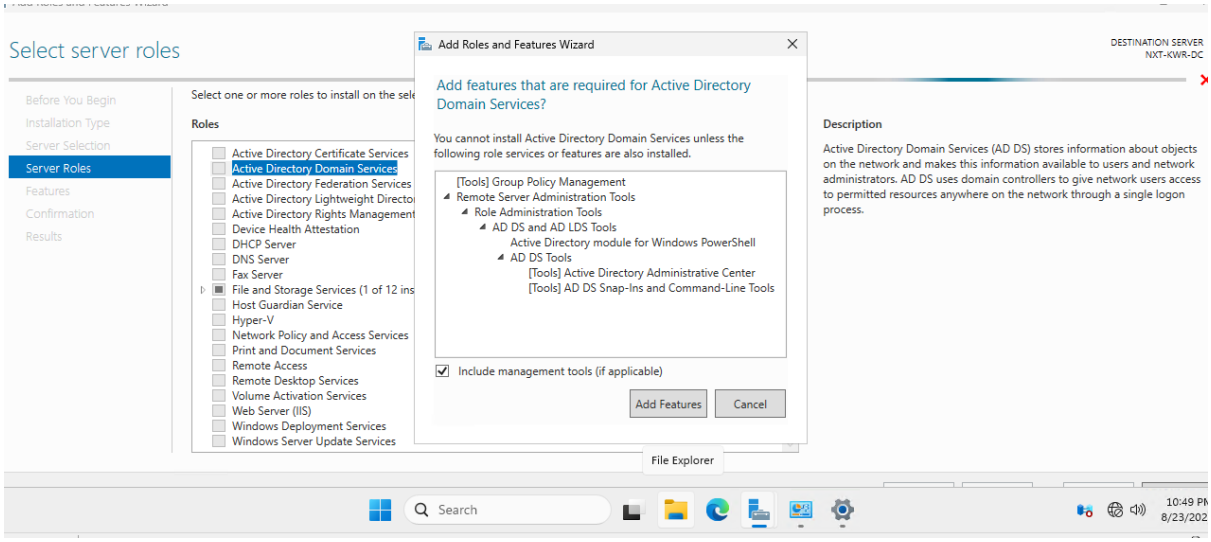
The Add Roles and Features Wizard provides two deployment methods. To configure an Active Directory Domain Controller, choose Option 1.

- Role-based or Feature-based Installation *(Selected)*
    - Function: Standard method used to add specific roles (such as AD DS, DNS, or DHCP) and features to a single server.
    - Purpose: The correct choice for building a Domain Controller in a lab environment.
- Remote Desktop Services Installation
    - Function: Used specifically for setting up Virtual Desktop Infrastructure (VDI), allowing multiple users to connect to a centralized workspace.
    - Purpose: Not required for basic Active Directory deployment.





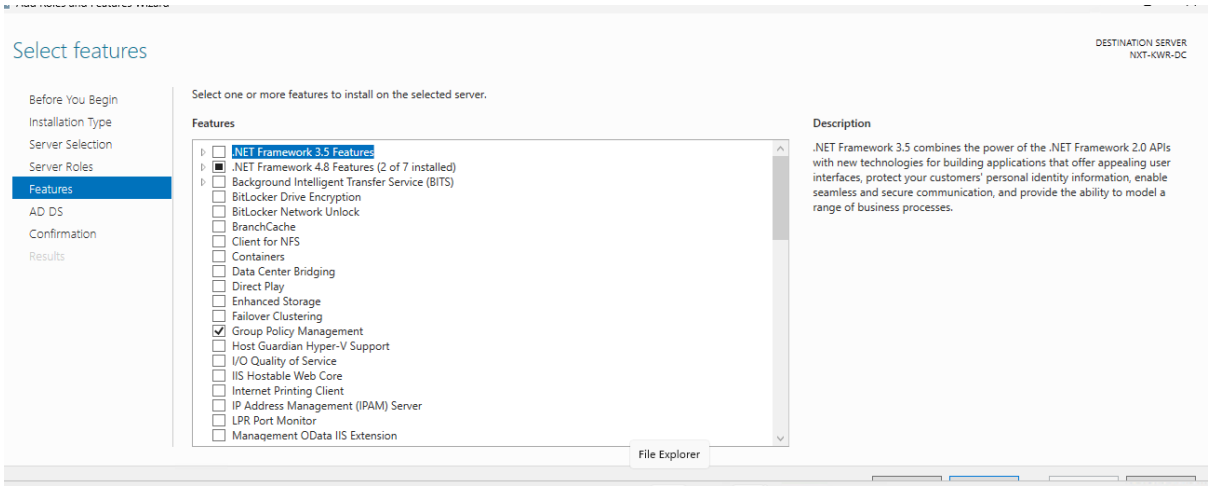
Then select Active directory domain service and click add features in pop-up box



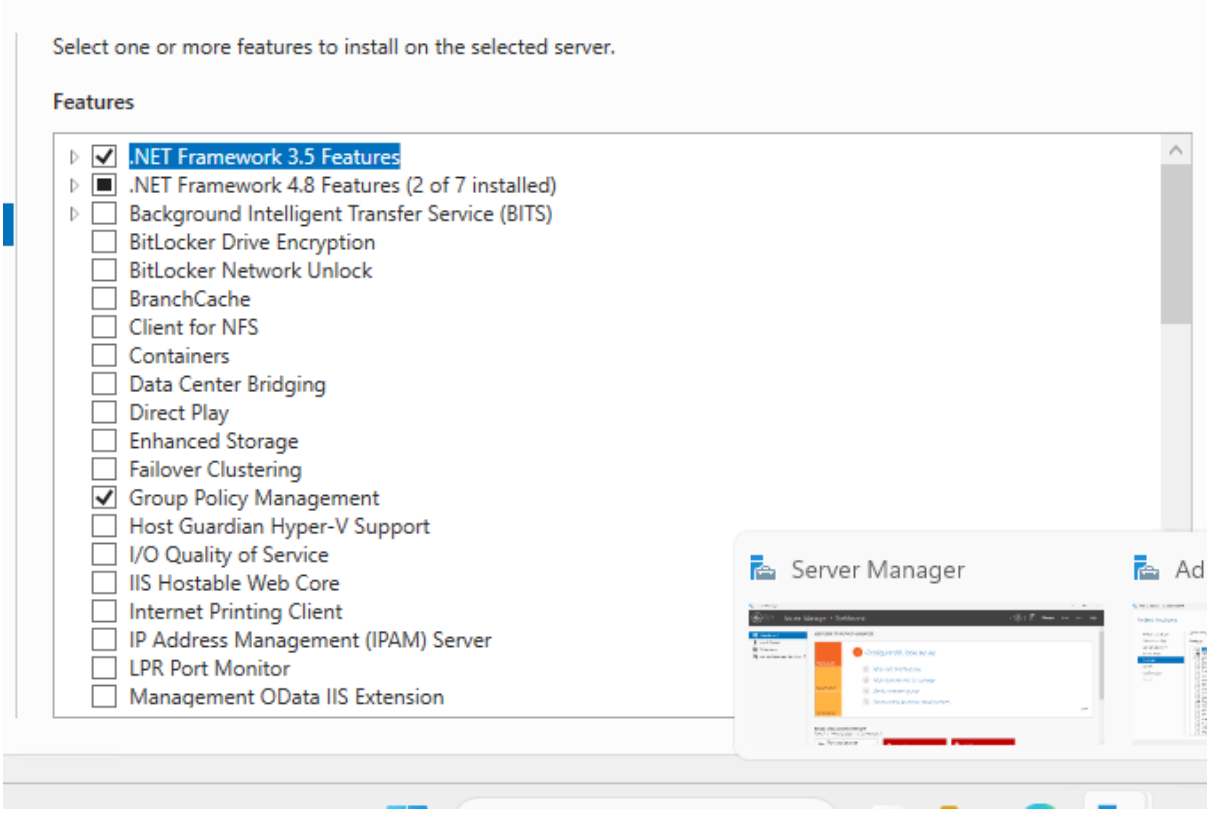
Phase: Selecting Features

At this stage, the wizard allows you to select optional software utilities that support primary server operations.

- **Understanding Features**
    - **Roles vs. Features:** While **Roles** define the core function of a server (e.g., Domain Controller), **Features** act as supporting tools, protocols, and software services.
    - **Automatic Selection:** When you install Active Directory Domain Services (AD DS), Windows automatically selects required dependencies like **Group Policy Management**.



We want to enable 3.5 feature for backward compatibility (ability to work with legacy system)



Troubleshooting: Missing Source Files for .NET 3.5

During the confirmation stage, the wizard displays a warning regarding missing source files for .NET Framework 3.5.

- **Issue:** Unlike standard features, .NET 3.5 files are not pre-cached on the virtual hard disk and must be retrieved directly from the original Windows Server ISO.
- **Resolution:** Mount the Windows Server ISO and specify its path in the wizard to supply the necessary installation files.

How to Specify the ISO Path in the Wizard

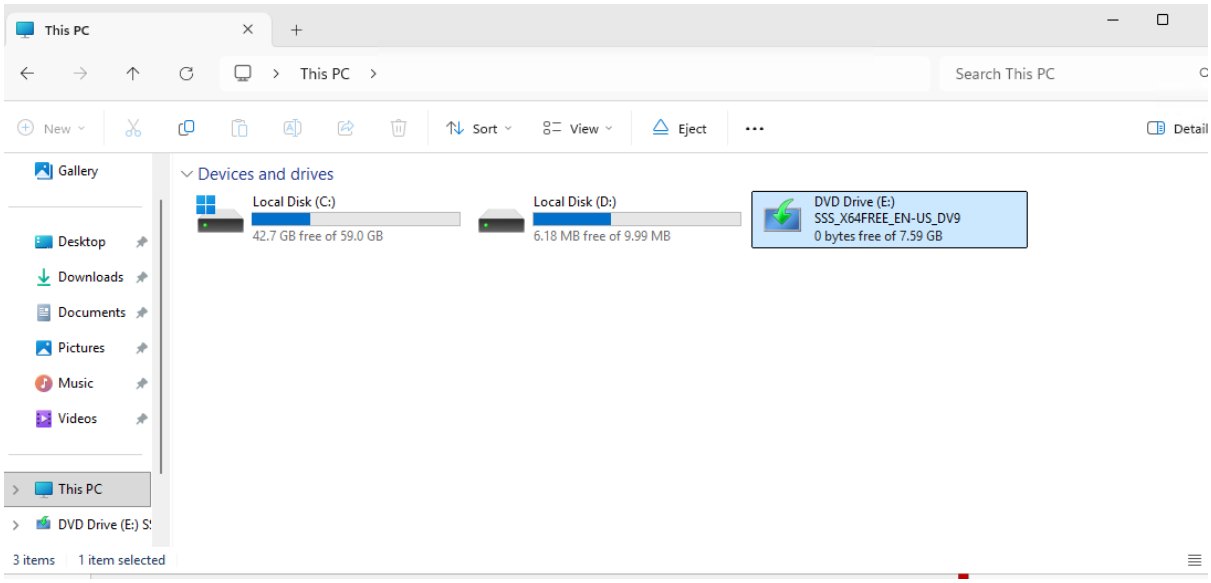
1. **Mount the ISO:** Attach the Windows Server ISO file to your virtual machine or host drive (e.g., Drive D:).
2. **Specify Alternate Source Path:** At the bottom of the **Confirmation** screen in the wizard, click **Specify an alternate source path**.
3. **Enter the Path:** In the **Path** box, enter the location pointing to the sxs folder inside the mounted ISO:

Plaintext

D:\sources\sxs

(Replace D: with the actual drive letter of your mounted ISO).

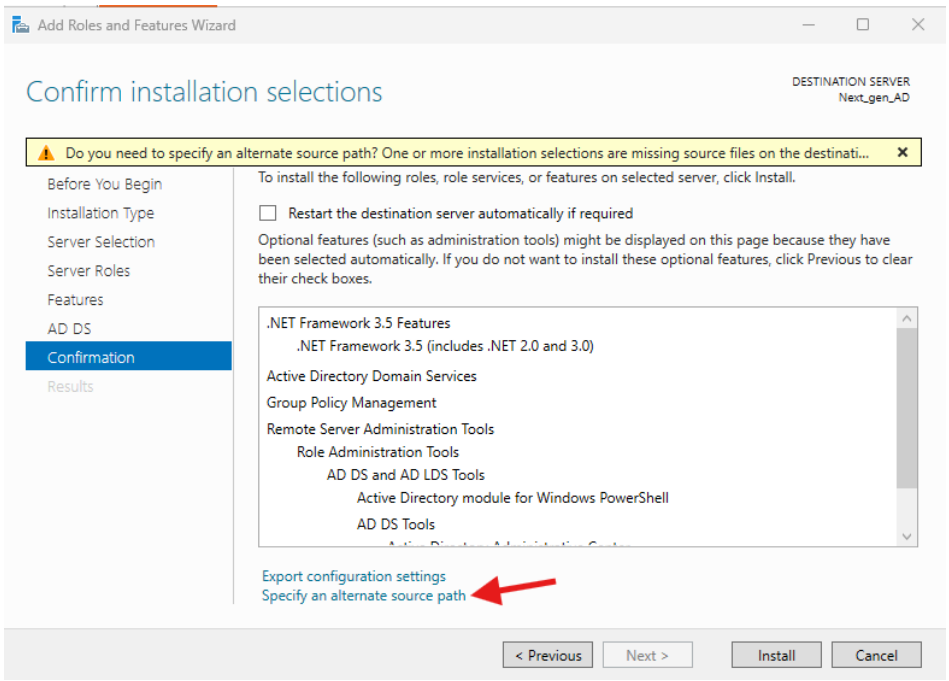
4. **Complete Installation:** Click **OK**, then proceed by clicking **Install**.

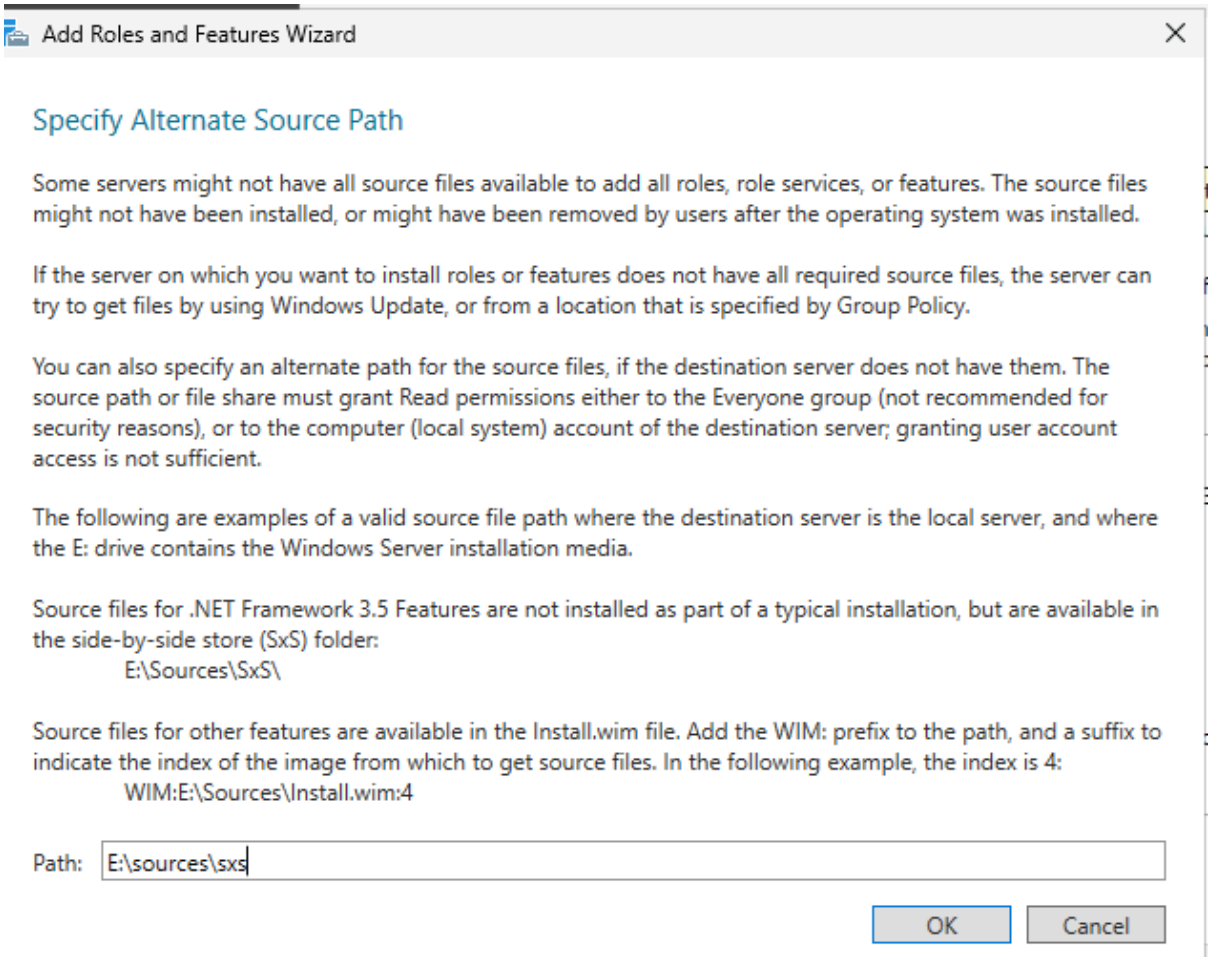


Click sources and click sxs file and copy the location:

E:\sources\sxs

Now we can add the file location manually

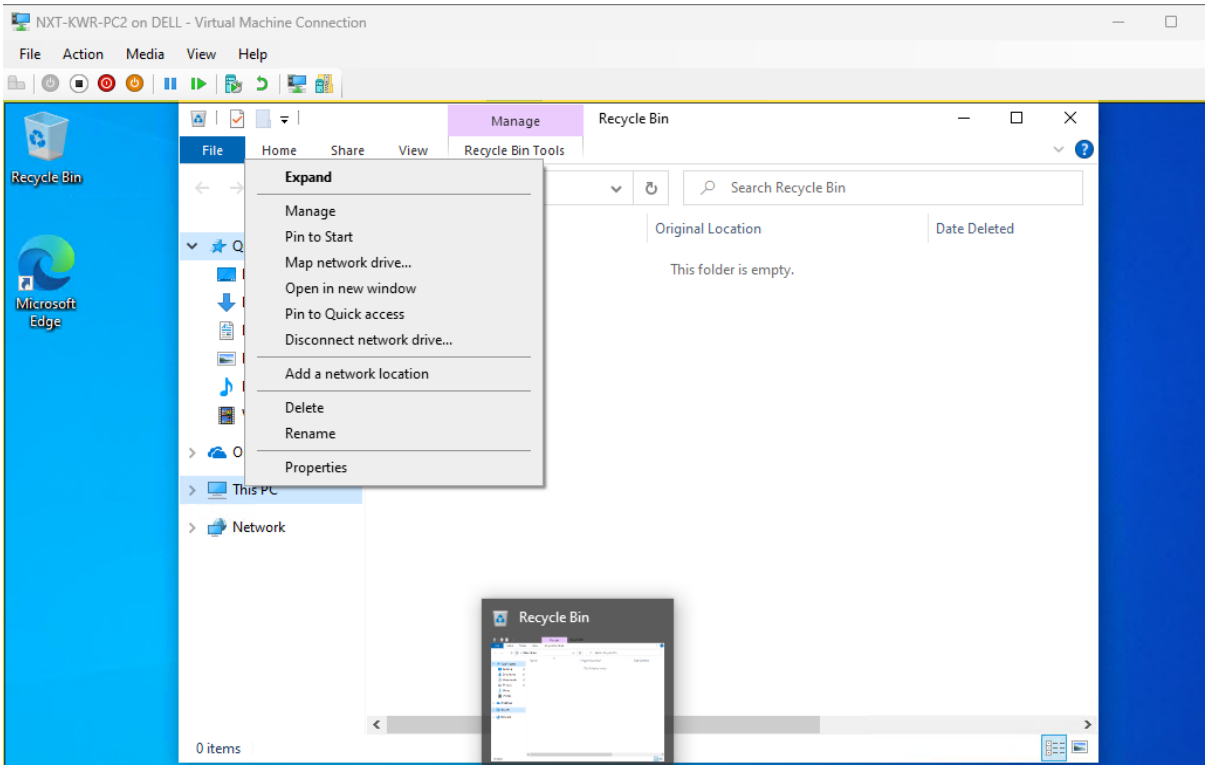




Then click install.

Next we setup the client computer

After preparing our Active Directory server, the next step is to set up the client computer, which is the machine that will later join the Active Directory domain.



The screenshot above shows a fresh client virtual machine, with basic Windows tools like Recycle Bin and Microsoft Edge visible on the desktop. This is the starting point, before we make any configuration changes.

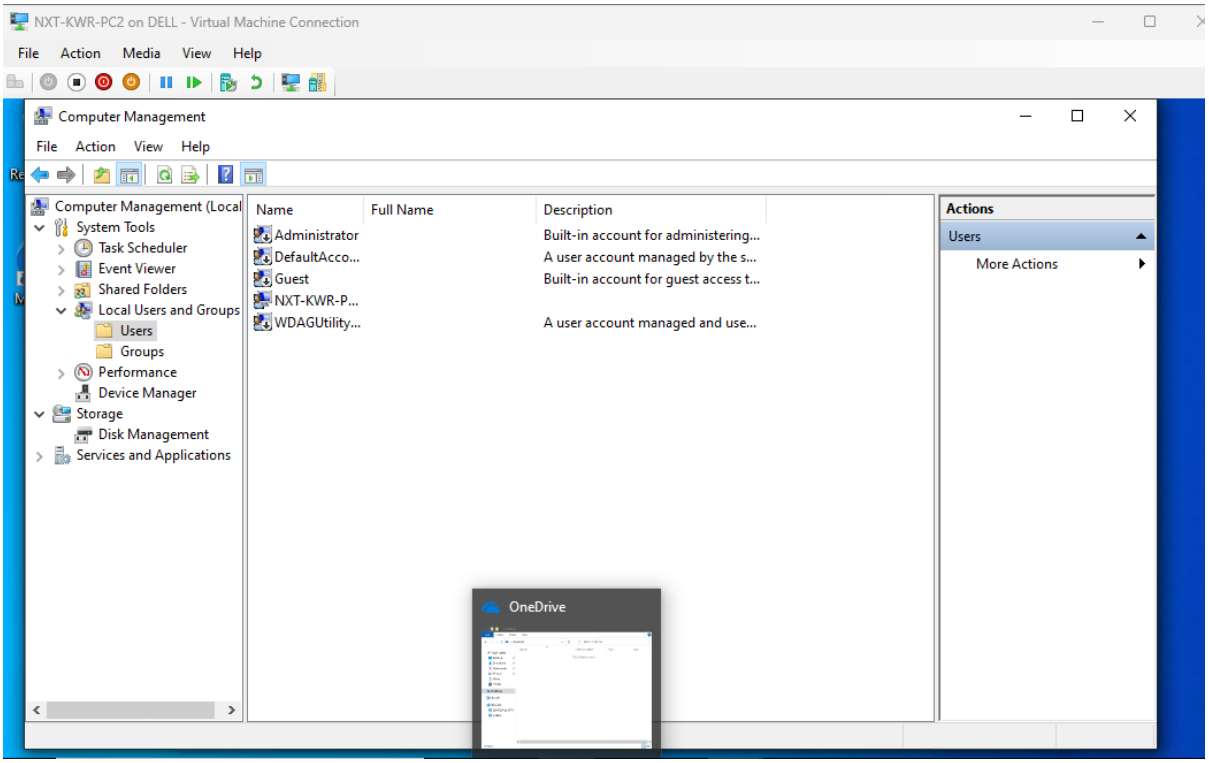
Windows Edition Limitation (Client Side)

Before going any further, there is an important limitation we need to be aware of. Microsoft restricts certain advanced management tools, including Local Users and Groups, to only the Windows Pro, Enterprise, and Education editions.

If our client machine happens to be running Windows Home edition, this feature will simply not be available, since Microsoft has deliberately disabled it on that edition. Even more importantly, Windows Home editions are not able to join an Active Directory Domain at all, no matter what we try.

Because of this, it is very important that we use Windows Pro, Enterprise, or Education edition for our client machine, and we should never attempt to use the Home edition for this kind of lab work.

Understanding the Default Local User Accounts



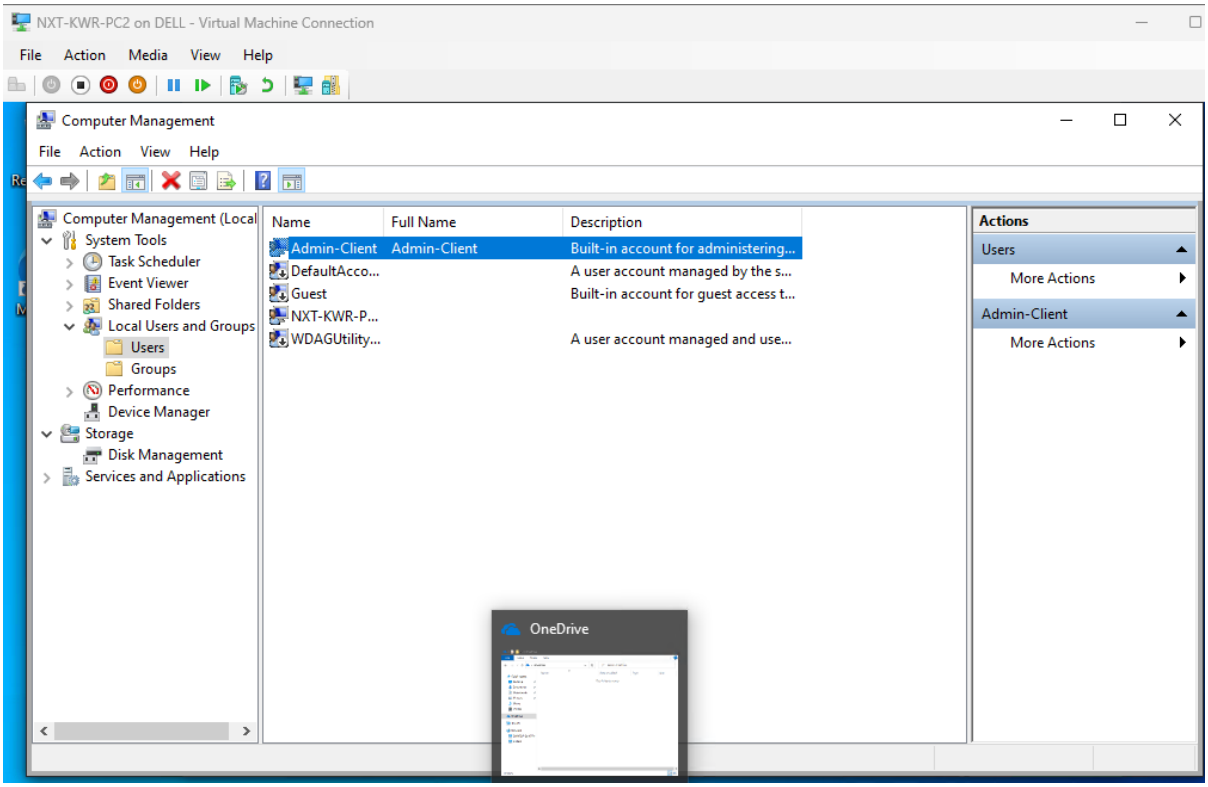
The screenshot above shows the Computer Management console, specifically the Local Users and Groups section, which lists all of the default accounts that exist on a freshly installed Windows machine.

- **Active Personal Account (for example, NXT-KWR-PC):** This is the primary user account that was created during the original Windows installation process. This account automatically belongs to the local Administrators group.
- **Built-in Administrator:** This is the master administrative account built into Windows. For security reasons, it is disabled by default (shown by a small down-arrow icon next to it in the list).
- **Guest:** This is a restricted account, meant only for temporary users, and it is also disabled by default.
- **System Accounts (Default Account, defaultuser0):** These accounts are managed automatically by the operating system itself, and are used for initial setup tasks and other internal system processes.
- **WDAGUtilityAccount:** This special account is used by a Windows feature called Windows Defender Application Guard, which runs untrusted web browsing sessions inside an isolated, virtualized container, in order to protect the main operating system from potential threats.

Enabling and Renaming the Administrator Account

The first thing we want to do on the client machine is enable the built-in Administrator account, set a password for it, and then rename it to something less obvious.

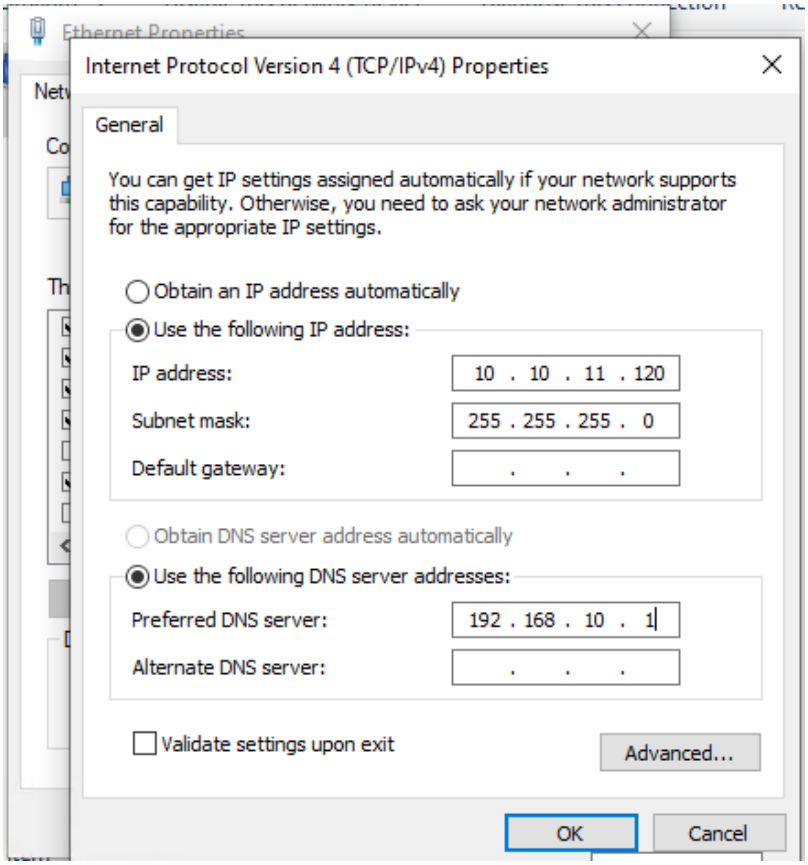
To do this, we right-click on the Administrator account, and select Properties, where we can enable the account. Then, we right-click on it again, and this time we can rename it, as well as add a password for it.



In the example above, the Administrator account has been renamed to "Admin-Client", and it is now shown as enabled in the list, ready to be used for managing this client machine.

Assigning a Static IP Address to the Client Computer

Just like we did for the AD server, we also need to assign a static IP address to the client computer. This ensures the client machine always keeps the same address, and can reliably find the Active Directory server on the network.



In the example above, the client computer is assigned the static IP address 10.10.11.120, with a subnet mask of 255.255.255.0.

Notice the Preferred DNS server field, which is set to 192.168.10.1. This is critical, because this address points directly to the Domain Controller, allowing the client machine to correctly perform Active Directory name resolution. Without this setting pointing to the correct DNS server, the client would not be able to find or join the domain later on.

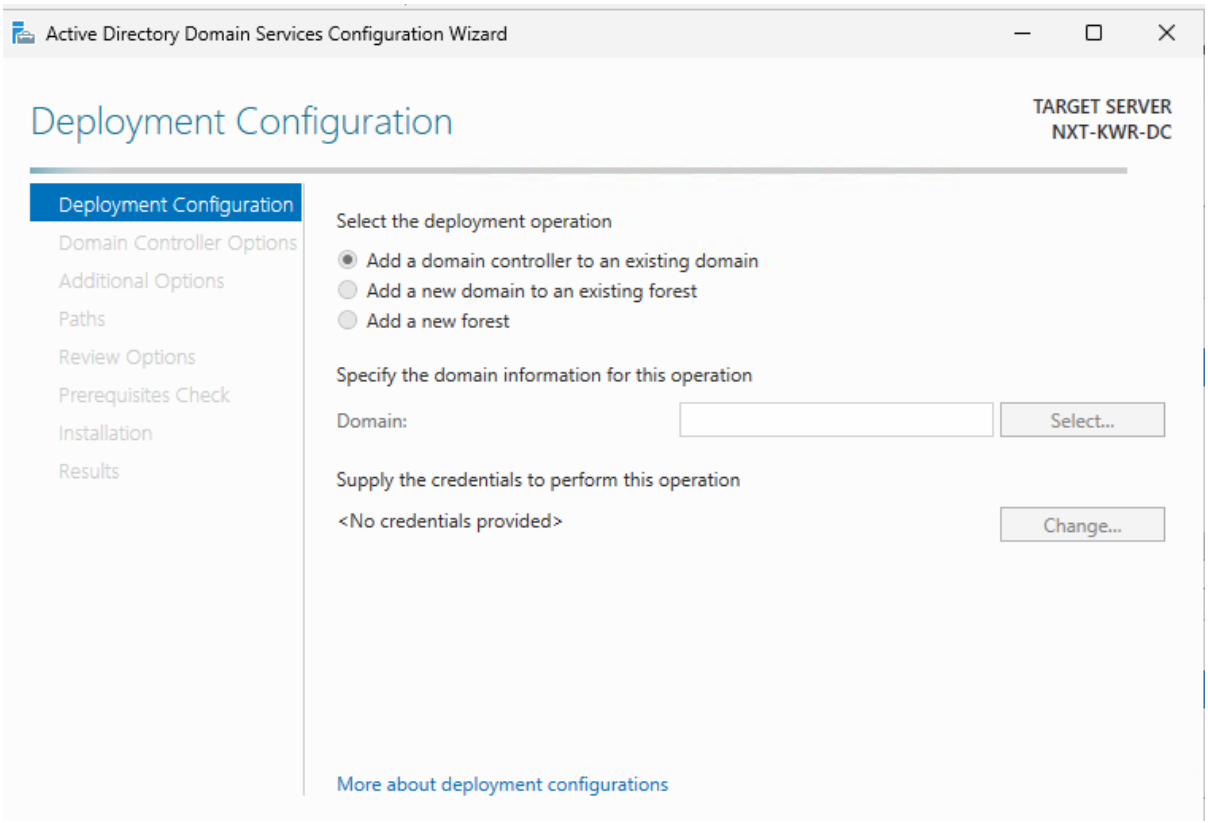
After setting the static IP address, we should also make sure to enable Remote Desktop on the client machine, so that we are able to manage it remotely if needed, just like we did with the server earlier.

Configuring the Active Directory Server (Promoting to a Domain Controller)

Now that both the server and the client computer are prepared, it is time to configure the Active Directory server itself, which involves a very important process.

An important note on terminology: when we install the Active Directory Domain Services role and then set up the very first domain, we do not call this "installing a domain." Instead, we say we are promoting the domain. Similarly, if we ever need to remove Active Directory later on, we say we are demoting the domain, not "uninstalling" it.

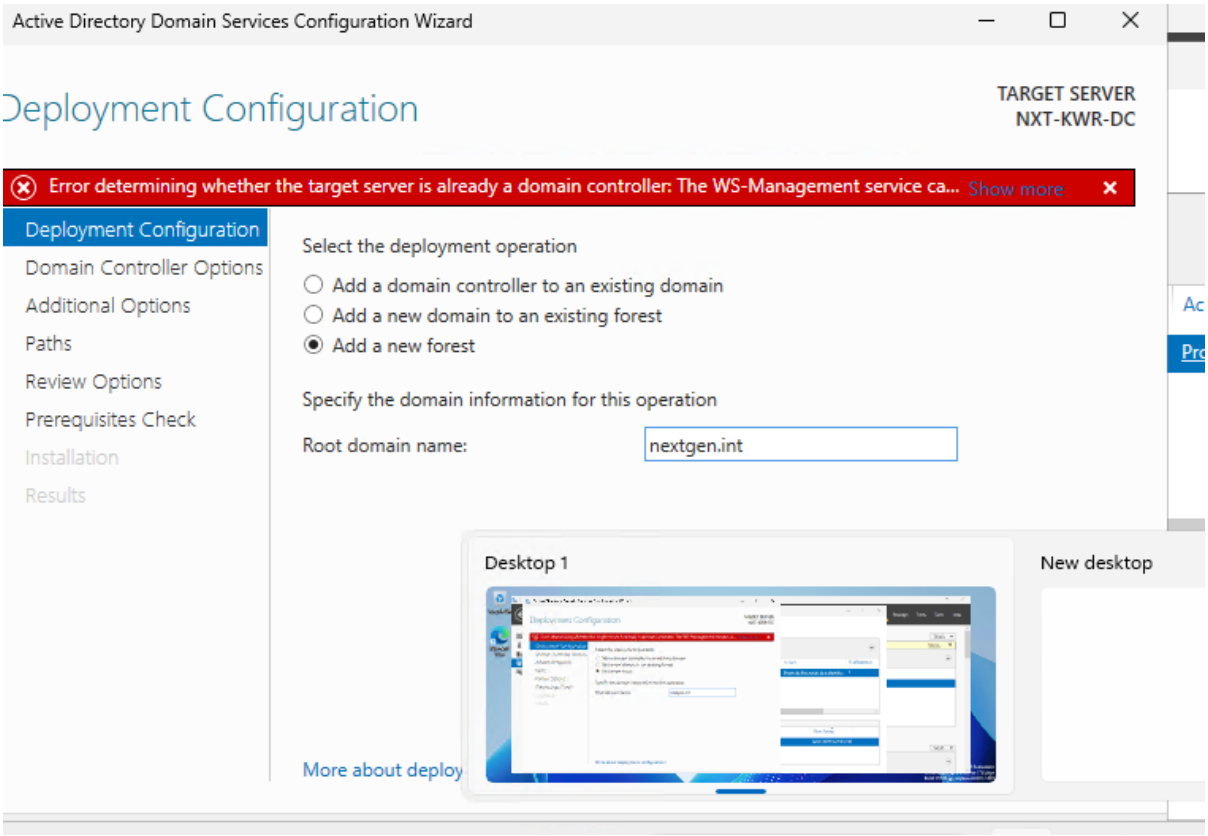
Deployment Configuration: Choosing the Right Option



When we launch the Active Directory Domain Services Configuration Wizard, we are presented with three main deployment options. Here is what each one means:

- **Add a domain controller to an existing domain:** This option is used to set up a backup or additional Domain Controller, within an environment where Active Directory is already running.
- **Add a new domain to an existing forest:** This option is used to create a child domain, or a tree domain, within an already established enterprise network.
- **Add a new forest:** This option is used when we are setting up the very first Domain Controller, for a brand-new network infrastructure that does not have Active Directory yet. This is the option we choose for our lab, since we are starting completely from scratch.

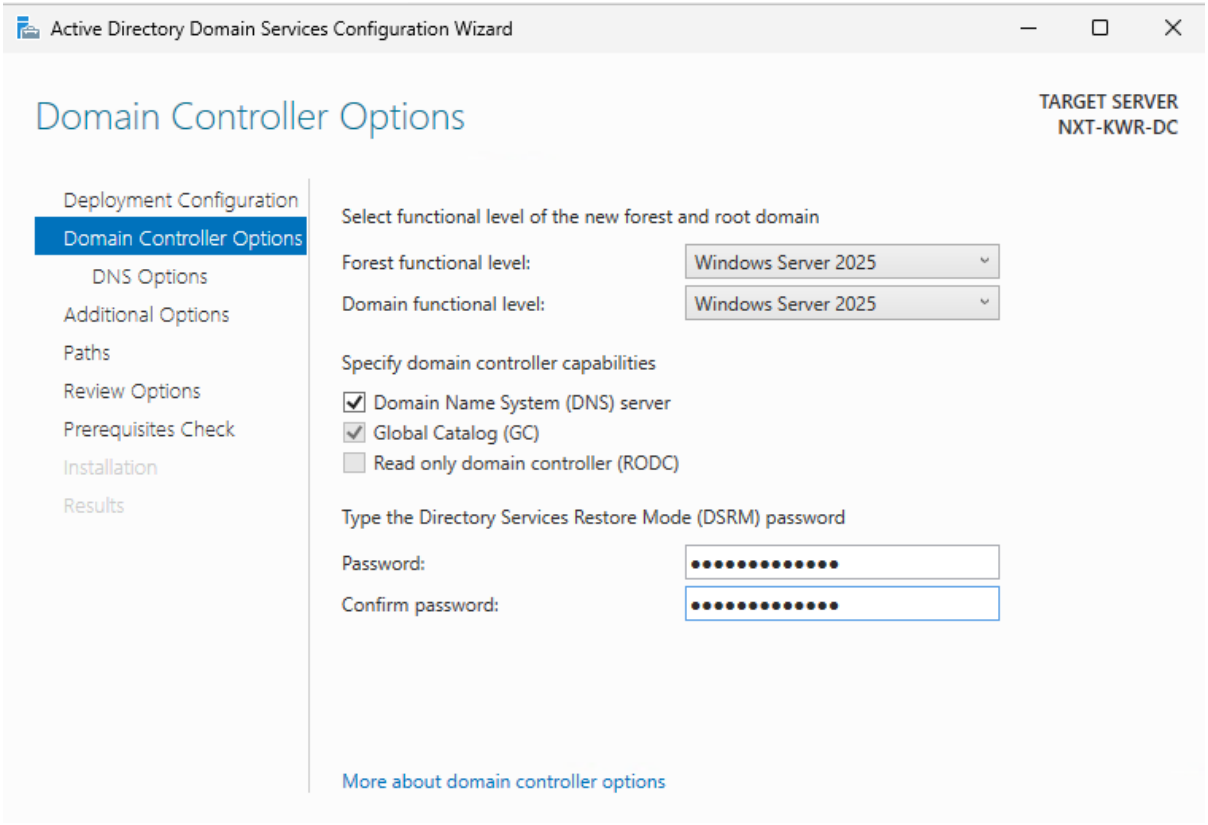
Choosing the Root Domain Name



Since we are creating a brand-new forest, we need to give it a root domain name. This should be a fully qualified name, for example: nextgen.int, nextgen.org, or nextgen.local.

In our example, we have chosen "nextgen.int" as our root domain name. It is worth noting that the warning message shown at the top of this screen, about not being able to determine whether the target server is already a domain controller, is completely normal at this stage, and can be safely ignored.

Domain Controller Options: Functional Levels



On this screen, we are asked to choose the Functional Level for both the forest and the domain. Here is what these settings actually mean:

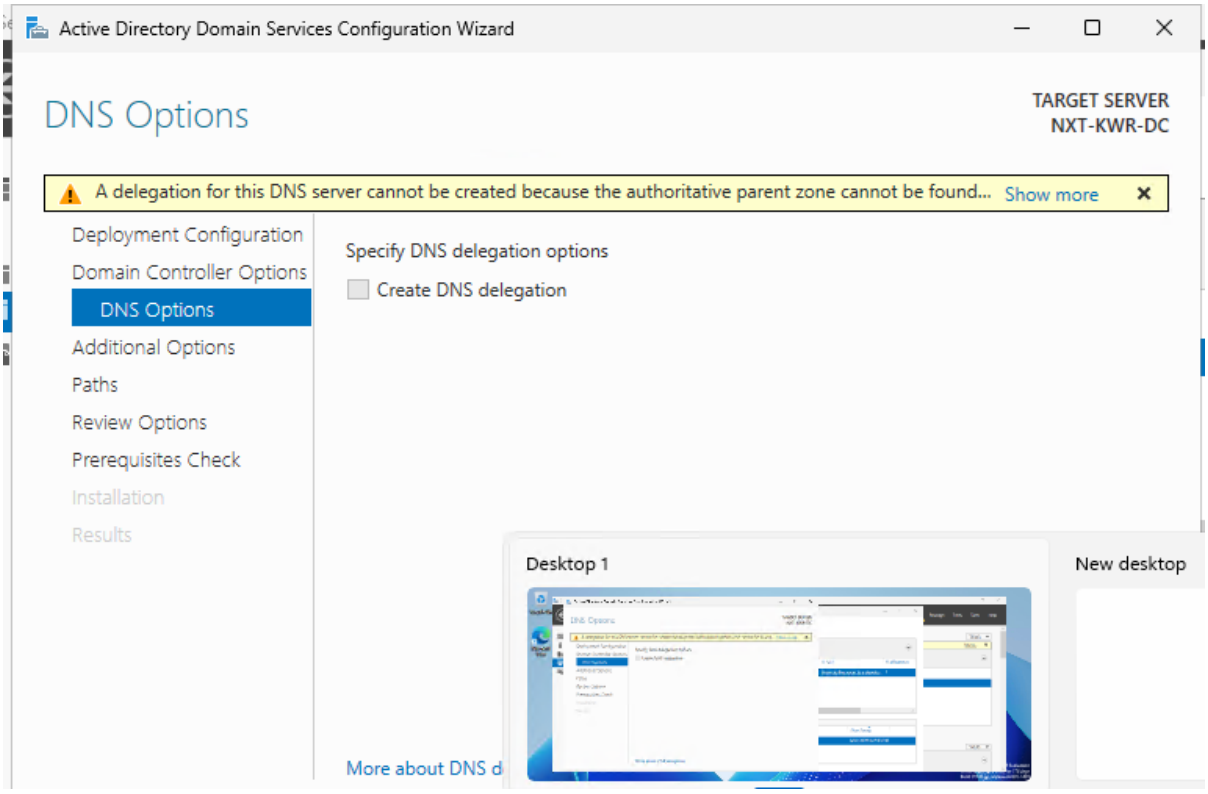
- **Forest Functional Level:** This setting decides the minimum Windows Server version that will be allowed, for any new Domain Controller added to the entire forest, in the future. By setting this to Windows Server 2025, we are locking the forest's features to the latest available standard.
- **Domain Functional Level:** This setting works the same way, but only applies to Domain Controllers within this specific domain.

The benefit of choosing Windows Server 2025 for both of these levels is that our environment gains access to the most advanced Active Directory security protocol upgrades, scalability improvements, and storage enhancements, which are exclusive to this modern OS version. It also prevents older, less secure servers, such as Server 2016 or 2019, from ever being added to our forest, since doing so could introduce older, legacy security vulnerabilities.

On this same screen, we also see a few additional core capabilities that are automatically selected for us:

- **DNS Server:** This is automatically installed alongside AD DS, since Active Directory relies heavily on DNS to manage name resolution, for the purpose of domain discovery.
- **Global Catalog (GC):** This is configured to host a searchable index of all the objects that exist across the entire directory database.
- **DSRM Password:** This is a special, dedicated recovery password, which is set specifically for something called Directory Services Restore Mode. This special mode is used to perform database maintenance, or disaster recovery, if something ever goes seriously wrong with Active Directory later on.

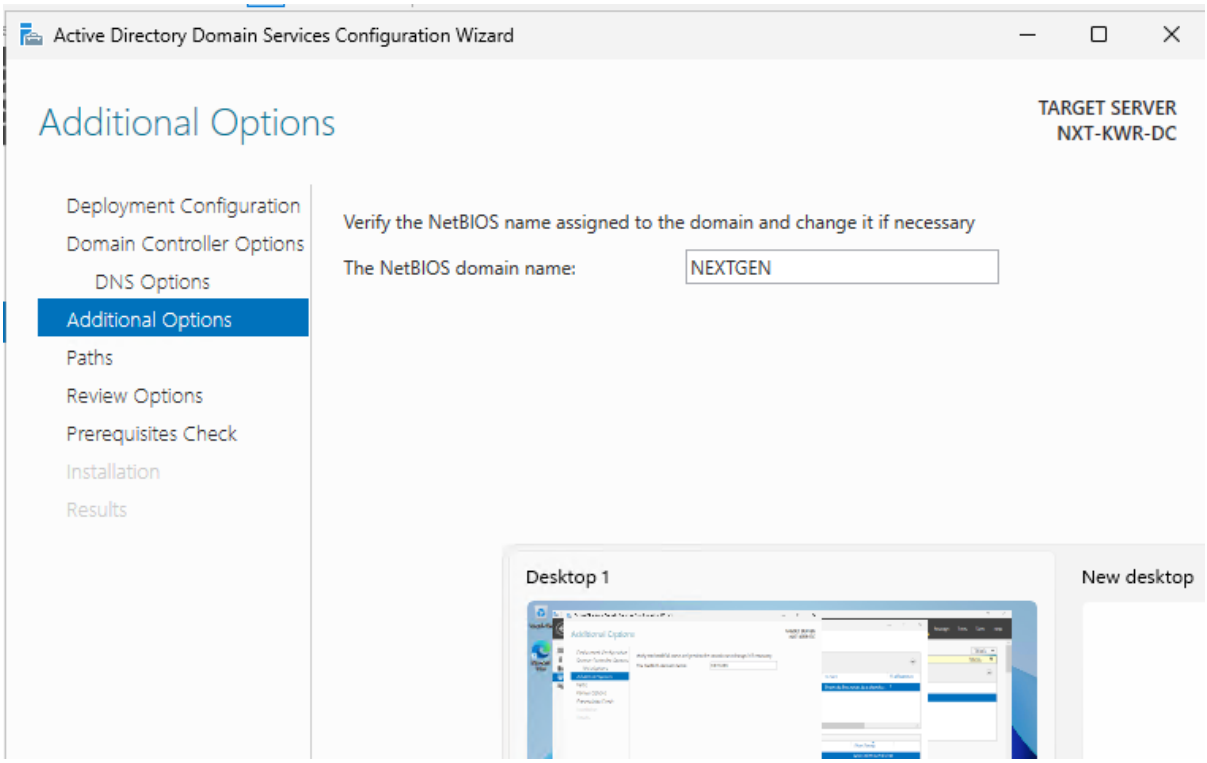
DNS Options: A Normal Warning



On this screen, we may see a warning message, saying that a delegation for this DNS server cannot be created, because the authoritative parent zone cannot be found.

This warning is completely normal and expected, whenever we are installing the very first, root domain of a brand-new forest, since there is no existing parent DNS zone for it to connect to. No action is required here; we simply leave the "Create DNS delegation" checkbox unchecked, and click Next to proceed.

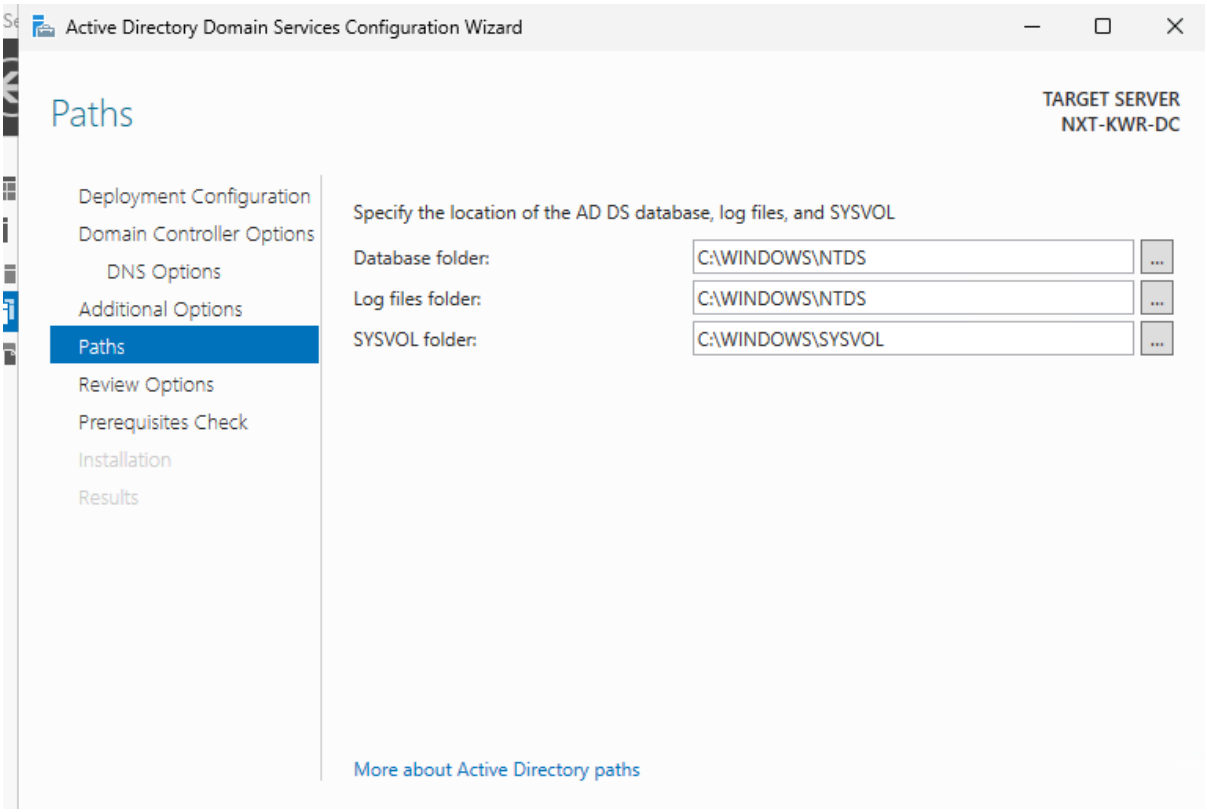
Additional Options: The NetBIOS Name



On this screen, the wizard automatically suggests a NetBIOS domain name for us, based on the domain name we entered earlier. In our case, since our domain is "nextgen.int", the suggested NetBIOS name is automatically set to "NEXTGEN".

The NetBIOS name is a shorter, older-style name for the domain, which is sometimes still used by legacy systems or certain applications, alongside the full modern domain name.

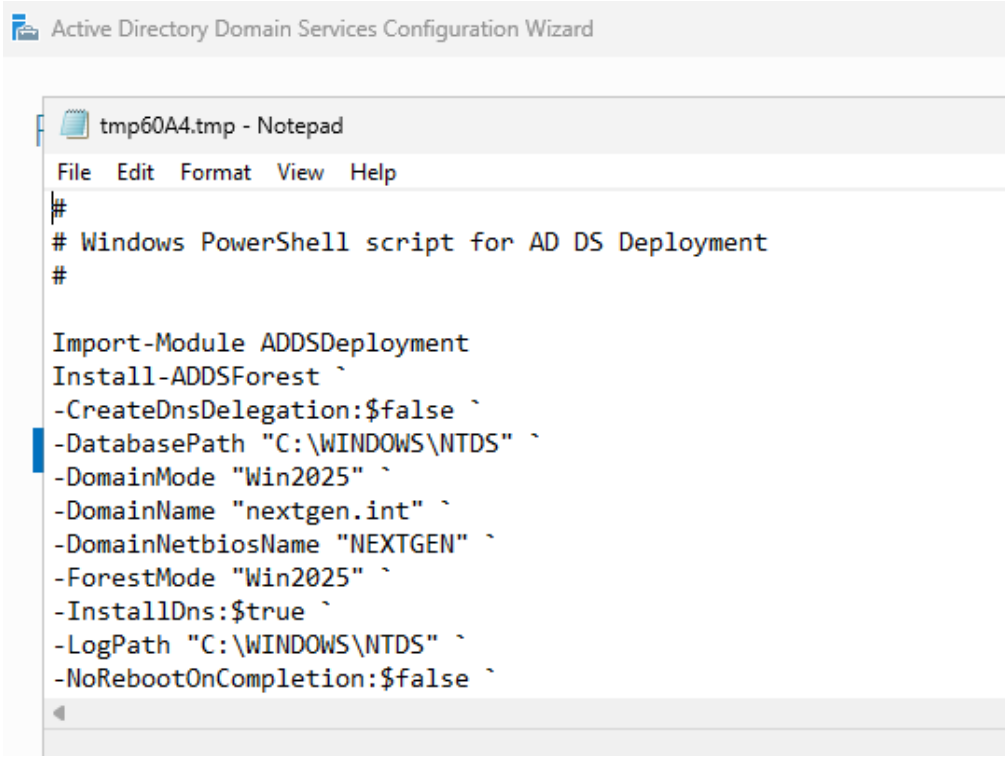
Paths: Where AD Files Are Stored



This screen lets us choose exactly where the Active Directory database folder, log files folder, and SYSVOL folder will be stored, on the server's hard drive.

In our lab, we simply leave these settings at their default locations. However, it is worth knowing that in some real-world, more advanced deployments, especially when using SQL Server or other heavy database systems, administrators sometimes do choose to move these folders to a different, dedicated storage location, for performance or organizational reasons.

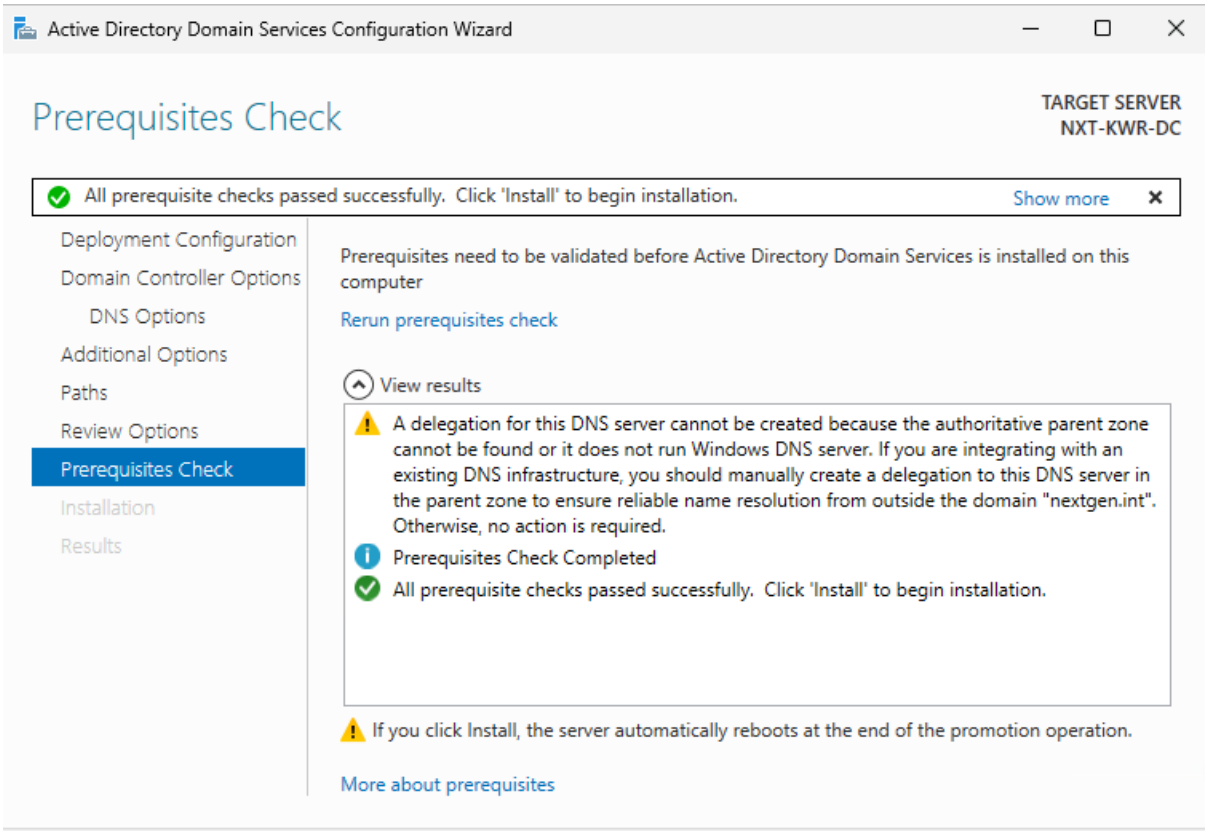
Review Options: Checking the Full PowerShell Script



Before the actual installation begins, the wizard shows us a full review of everything we have selected so far. Very usefully, it also shows us the exact PowerShell script that corresponds to all of these settings.

This is extremely helpful, because it means that instead of manually clicking through this graphical wizard every single time, an administrator could simply copy this PowerShell script, and run it directly, to automate the exact same Active Directory installation on another server in the future.

Prerequisites Check: Confirming Everything Is Ready

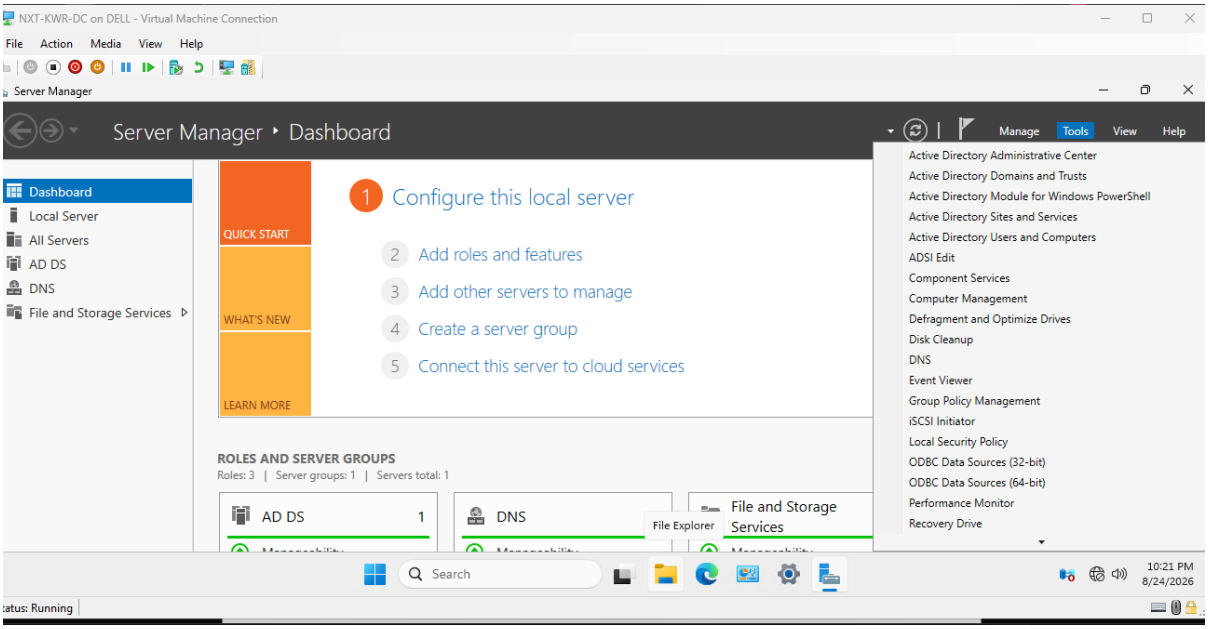


Before we are allowed to click the Install button, the wizard automatically runs a full prerequisites check, to confirm that our server meets all the requirements needed to become a Domain Controller.

As shown in the screenshot above, once this message appears saying "All prerequisite checks passed successfully," we are safely able to click the Install button, to begin the actual installation process.

It is important to remember that once we click Install, the server will automatically restart on its own, once the promotion process is fully completed.

After Installation: Exploring the Newly Promoted Server

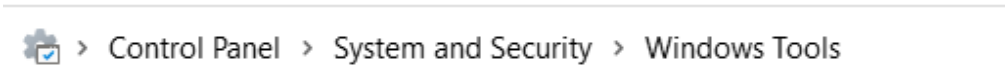


After the server has finished installing and restarting, we can open Server Manager to see our new Active Directory environment. Notice that under "Roles and Server Groups," we can now clearly see AD DS and DNS listed as active roles running on this server.

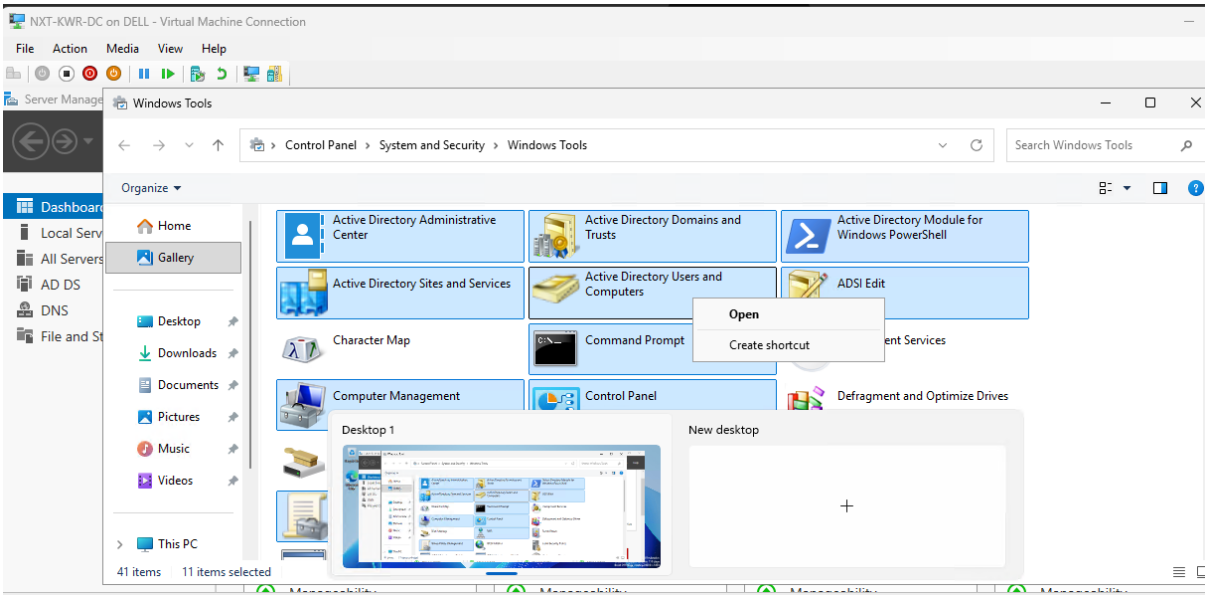
Also notice the Tools menu, shown open on the right side of the screenshot. This menu now contains many new, Active Directory-specific management tools, such as Active Directory Administrative Center, Active Directory Domains and Trusts, Active Directory Sites and Services, and most importantly, Active Directory Users and Computers, which we will use very often, to manage our users and computers going forward.

Adding Useful Shortcuts to the Desktop

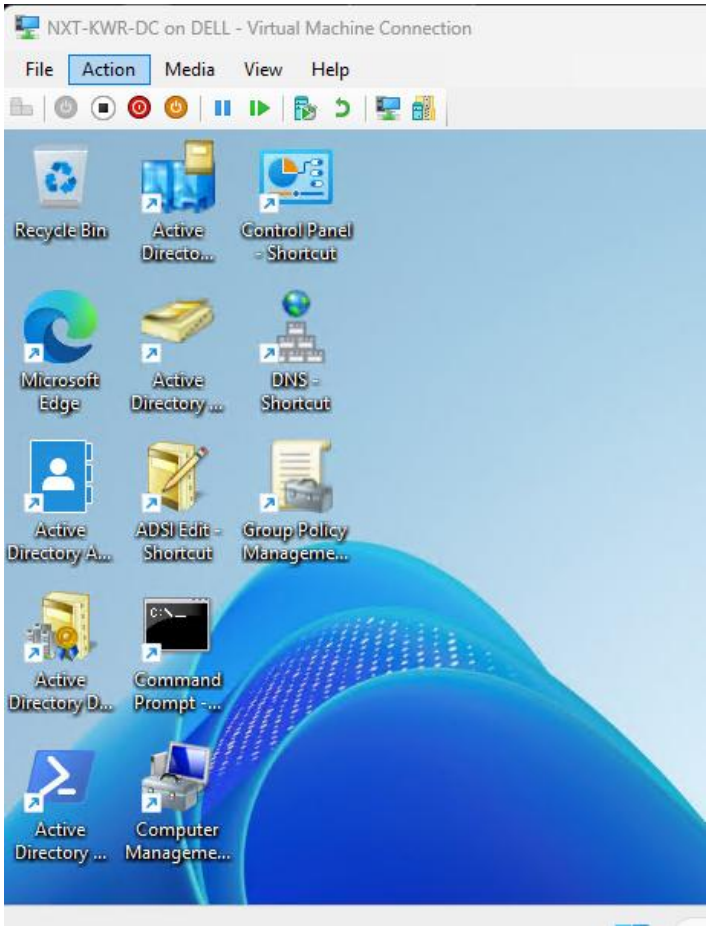
Since we will be using these Active Directory tools very frequently, it makes sense to create convenient desktop shortcuts for them, instead of navigating through the Tools menu every single time.



These tools can also be found by navigating through Control Panel, then System and Security, then Windows Tools, as shown in the path above.



In the screenshot above, we can see the full Windows Tools folder, containing all of the available management tools, including Active Directory Administrative Center, Active Directory Users and Computers, Computer Management, and more. By right-clicking on any of these tools, we can select "Create shortcut," to place a convenient shortcut directly on our desktop.



The screenshot above shows the final result: a desktop with several useful shortcuts already created, including Active Directory Users and Computers, Control Panel, DNS, ADSI Edit, Group Policy